
INDIAN STATISTICAL INSTITUTE, KOLKATA

Bachelor of Statistics



— STATISTICS COMPREHENSIVE —

The Digital Dividend in Informal Markets

*A Statistical Analysis of UPI Adoption
Among Street Vendors on the B.T. Road Corridor*

MAY 2026

Class of 2026

Advisor & Instructor

PROF. DEBASIS SENGUPTA
APPLIED STATISTICS UNIT

PROF. SUMAN MAJUMDAR
INTERDISCIPLINARY STATISTICAL RESEARCH UNIT

Indian Statistical Institute, Kolkata

Contents

I	Introduction and Background	6
1	Background and Motivation	7
1.1	The Indian Digital Payments Revolution	7
1.2	Study Area: The B.T. Road Corridor	8
1.3	Importance of Micro-Level Evidence	8
1.4	Research Gap and Need for the Study	9
2	Project Objectives	10
3	Fieldwork Practices and Respondent Interaction	12
II	Survey Design and Data	14
4	Survey Design	15
4.1	Design Philosophy	15
4.2	Questionnaire Structure	16
4.3	Sampling Strategy	17
5	Data Pre-Processing	19
5.1	Data Cleaning	19
III	Descriptive and Exploratory Analysis	21
6	Demographic Profile of Street Vendors	22
6.1	Gender Distribution in the Vending Sector	22
6.2	Age Group Analysis and Workforce Maturity	23
6.3	Type of Goods	23
6.4	Analysis of Market Longevity	24
6.5	Operational Insights:	24
6.6	Cross-Tabulation Analysis:	25
6.7	Generational Operational Trends:	25

7	The UPI Ecosystem	28
7.1	UPI Acceptance Among Street Vendors	28
7.2	UPI Adoption Reasons	29
7.3	UPI Rent and Cost Structure	29
7.4	UPI Transaction Frequency per 10 Customers	30
7.5	Network Issue Frequency Among Street Vendors	30
8	Analysis of Business Earnings and Payment Trends	32
8.1	Analysis of Ticket Size Distribution	32
8.2	Analysis of Earnings by Business Trajectory	33
8.3	Analysis of Earnings Distribution (Histogram)	35
8.4	Analysis of Highest Earnings by Business Trajectory	36
8.5	Analysis of Highest Earnings Distribution (Histogram)	37
8.6	Analysis of UPI Paying Customers (Last Hour)	38
8.7	Analysis of Total Customers (Last Hour)	39
8.8	Analysis of Average Customers per Day	40
8.9	Analysis of Perception: Does UPI Help Save More?	41
8.10	Analysis of Payment Method Distribution	42
9	Cash, UPI, and Behavioural Dynamics Among Street Vendors	43
9.1	Analysis of Preferred Mode of Transaction	43
9.2	Analysis of Customers Lost Due to Lack of Cash	44
9.3	Analysis of Concern Regarding Digital Records	45
9.4	Analysis of Cash Usage Among the Last 10 Customers	47
9.5	Analysis of Transaction Speed Using UPI	48
9.6	Analysis of Customers Lost Due to Lack of UPI	49
10	Data at a Glance	50
IV	Inferential Analysis	53
11	UPI Intensity Index	54
11.1	Reliability-Weighted Bayesian Estimation of UPI Intensity	54
11.1.1	Observed Data Structure	54
11.1.2	Model Specification	55
11.1.3	Reliability Weighting via Internal Consistency	55
11.1.4	Posterior Estimator	56
11.1.5	Advantages of the Bayesian Estimator	56
11.1.6	Illustrative Example	57

12 Revenue Volatility Index	58
12.1 Bayesian Estimation of Revenue Instability	58
12.1.1 Observed Logarithmic Revenue Range	58
12.1.2 Likelihood and Prior Specifications	58
12.1.3 Empirical Initialisation of Hyperparameters	59
12.1.4 Posterior Distribution and the Instability Index	59
12.2 Theoretical and Empirical Justification of the Bayesian Instability Index	59
12.2.1 Comparison with the Parkinson Volatility Estimator	60
13 UPI Adoption	65
13.1 Motivation and Objective	65
13.2 Bayesian formulation of UPI intensity	65
13.3 Questionnaire variables considered	66
13.4 Analysis	67
13.5 Customer Arrival across UPI usage	72
13.6 Summary of findings	74
14 Cash Change Shortage and UPI Adoption	75
14.1 Motivation and Objective	75
14.2 Variable Construction and Data Preparation	75
14.3 Revenue Loss Monetisation	76
14.4 Descriptive Results	76
14.5 UPI Intensity as a Mitigation Mechanism	79
14.6 Regression Framework	80
14.7 Regression Results	81
14.8 Interpretation	82
14.9 Overall Conclusion	82
15 Daily Revenue Volatility and UPI Adoption	84
15.1 Motivation and Objective	84
15.2 Revenue Volatility Index Used in the Analysis	84
15.3 Data and Descriptive Statistics	86
15.4 Exploratory Plots	87
15.5 Hypothesis Testing	88
15.5.1 Bivariate Relationship	89
15.5.2 Sensitivity Analysis	89
15.6 Overall Conclusion	89
16 Vendor Perception of UPI	90
16.1 Motivation and Objective	90

16.2 Variables and Statistical Approach	90
16.3 Transaction Speed	91
16.4 Savings Behaviour and Financial Tracking	91
16.5 Digital Trust, Network Issues, and Mode Preference	92
16.6 Reliability and Business Experience	93
16.7 Overall Conclusion	94
17 Motivations behind UPI Adoption	95
17.1 Motivation and Objective	95
17.2 Primary Motivating Factors	95
17.3 Business Duration and Adoption Motives	96
17.4 UPI Fee Arrangements	97
17.5 Fee Arrangement and Business Trajectory	98
17.6 Pre-model Screening for Fee Arrangement	98
17.7 Age and UPI Adoption Status	99
17.8 UPI Adoption Utility	99
17.9 Overall Conclusion	100
V Conclusions	101
18 Conclusions	102
19 Acknowledgement	105
VI Appendix	106
A Appendix	107

Part I

Introduction and Background

Chapter 1

Background and Motivation

1.1 The Indian Digital Payments Revolution

Over the last decade, India has witnessed a remarkable transformation in its payment ecosystem. Traditionally, retail transactions in the country were overwhelmingly cash-driven, particularly within the informal sector where roadside traders, small vendors, and daily market sellers operated largely outside formal banking networks. The introduction of the Unified Payments Interface (UPI) by the National Payments Corporation of India (NPCI) in April 2016 marked a significant turning point in this landscape by enabling fast, low-cost, and interoperable digital transactions through smartphones.

Unlike earlier digital payment systems that depended heavily on banking infrastructure, debit cards, or point-of-sale machines, UPI simplified the process of money transfer by allowing direct bank-to-bank transactions through mobile applications and QR codes. This substantially reduced both the technological and financial barriers associated with digital payments. The rapid spread of affordable smartphones, inexpensive mobile internet services, and user-friendly payment applications further accelerated adoption across different socio-economic groups.

The expansion of UPI became particularly prominent following the demonetisation policy implemented in November 2016. The temporary shortage of physical currency encouraged consumers and merchants alike to experiment with alternative payment mechanisms. Over time, what began as a short-term adjustment gradually evolved into a lasting behavioural change. Digital transactions became increasingly integrated into everyday commercial activities, extending far beyond organised retail markets into

neighbourhood shops, tea stalls, fruit vendors, and roadside businesses.

Today, QR-code payment boards are a common feature even among small informal vendors operating with minimal capital investment. This shift reflects not only technological diffusion but also changing consumer preferences. Urban customers increasingly prefer convenient and cashless modes of payment, particularly in high-mobility environments where speed and transaction efficiency are important.

1.2 Study Area: The B.T. Road Corridor

The present study focuses on the Barrackpore Trunk Road (B.T. Road) corridor, one of the most commercially active transport routes in the northern region of Kolkata and its adjoining urban areas. Stretching from Sinthi More to Dunlop Crossing, the corridor passes through densely populated localities including Bonhooghly, Gopal Lal Tagore Road (commonly referred to as GLT Road), Dunlop Bridge area, Baranagar, and adjoining commercial zones along the B.T. Road corridor. The region experiences continuous movement of commuters, office workers, students, industrial labourers, and local residents throughout the day, thereby creating a highly dynamic retail environment.

B.T. Road contains a large concentration of informal-sector vendors operating in both permanent and semi-permanent forms. These vendors sell a diverse range of products including tea, snacks, fruits, vegetables, garments, footwear, mobile accessories, hardware goods, and various household items. Since many of these businesses rely heavily on daily pedestrian traffic and high-frequency transactions, transaction efficiency becomes particularly important in determining their earnings and customer retention.

A preliminary recce survey conducted prior to the main data collection indicated that digital payment acceptance has become nearly universal among vendors in several parts of the corridor. QR-code payment systems were observed across a wide variety of businesses irrespective of their scale of operation. However, considerable variation remains in the intensity of UPI usage.

1.3 Importance of Micro-Level Evidence

The rapid expansion of digital payment systems in India has generated extensive discussion regarding financial inclusion, technological modernisation, and the transition toward a less cash-dependent economy. However, much of the available evidence is derived from aggregate indicators such as transaction volumes, banking statistics, or platform-level adoption figures. While such measures successfully capture the scale of

digital payment growth, they do not adequately explain how these transformations affect the everyday economic realities of individuals operating within the informal sector.

Street vendors typically function under conditions of considerable economic uncertainty. Their earnings often depend on daily customer flow, local demand fluctuations, weather conditions, and seasonal variation. Unlike formal businesses, most vendors possess limited savings mechanisms and restricted access to institutional credit that could help stabilise income during periods of low sales.

Micro-level evidence becomes important because it allows direct examination of practical economic effects such as reduction in transaction friction, customer retention, and revenue stability. UPI transactions potentially reduce problems arising from shortage of change, delayed cash handling, and failed sales due to inability to process exact payments.

1.4 Research Gap and Need for the Study

Although digital payments have become an important component of India's economic transformation, comparatively little attention has been devoted to understanding their impact on small informal vendors at the local level. Existing studies frequently concentrate on macroeconomic trends, banking infrastructure, technological adoption, or consumer behaviour, while the economic experiences of roadside vendors remain underrepresented in quantitative research.

Most available discussions regarding digital payment usage among street vendors are descriptive in nature and rely heavily on anecdotal observations. There remains limited statistical evidence examining whether UPI adoption actually improves outcomes such as sales consistency, reduction in transaction losses, customer expansion, or revenue stability.

The present study attempts to address this gap by constructing and analysing a primary dataset collected from vendors operating along the B.T. Road corridor. Through this analysis, the study seeks to examine the relationship between digital payment usage and indicators of economic stability within the informal retail sector.

Chapter 2

Project Objectives

The rapid expansion of UPI-based transactions among informal-sector vendors has substantially altered the nature of day-to-day retail exchange in urban markets. However, the economic consequences and behavioural implications of this transition remain insufficiently examined at the micro level. The present study therefore seeks to investigate how varying levels of UPI usage influence customer flow, transaction efficiency, revenue stability, and vendor behaviour along the B.T. Road corridor.

The major objectives of the study are as follows:

1. **Customer Volume and UPI Intensity**

To examine whether there exists a statistically significant difference in the number of customers served between vendors with low UPI intensity and those with high UPI intensity.

2. **Sales Loss Due to Change Shortage**

To estimate the magnitude of sales losses arising from the cash change-shortage problem and to investigate whether UPI adoption helps mitigate such transaction-related losses.

3. **Revenue Volatility and Digital Transactions**

To estimate short-term daily revenue volatility among vendors and determine whether higher UPI intensity is associated with lower revenue volatility after controlling for factors such as goods type and business location.

4. **Vendor Perception and Behavioural Response**

To conduct an opinion survey examining vendor perceptions regarding transaction speed, financial record-keeping, savings behaviour, and trust in digital payment systems, with responses analysed across goods category, age group, and mode of operation.

5. **Drivers of UPI Adoption**

To identify the principal motivating factors behind UPI adoption among vendors, including customer demand, competitive pressure, convenience, and operational considerations, along with the structural fee arrangements under which vendors conduct digital transactions.

Chapter 3

Fieldwork Practices and Respondent Interaction

Since the study involved direct interaction with informal-sector vendors and included questions related to earnings, transaction practices, and financial behaviour, the survey process was designed to encourage voluntary participation, respectful communication, and respondent comfort during data collection. As this was an academic student project rather than a formally regulated human-subject study, these practices were incorporated into field interactions to support transparent and considerate engagement with respondents. Many vendors operate without formal institutional protection and may be hesitant to disclose business-related information. The survey was therefore designed in a manner that prioritised respondent comfort, privacy, and informed participation.

Interaction Preamble

Before beginning each interaction, respondents were briefly informed about the academic purpose of the study and that participation was entirely voluntary. Rather than following a fixed script, these points were communicated naturally within conversation so that respondents could engage comfortably and ask questions where necessary. The intention was to maintain transparency while keeping the interaction informal and non-intrusive.

The following fieldwork practices were followed during survey:

1. **Voluntary Participation**

All respondents were clearly informed that participation in the survey was entirely

voluntary. Vendors were free to refuse any question or discontinue the interview at any stage without any consequence.

2. **Careful Handling of Income-Related Questions**

Since questions regarding earnings can be sensitive for informal-sector vendors, respondents were informed that the information would be used strictly for academic purposes only. Income-related questions were asked respectfully and in approximate or range-based form wherever appropriate in order to minimise discomfort and encourage honest responses.

3. **Protection of Identity and Privacy**

No sensitive personal information such as Aadhaar number, bank account details, or phone number was collected during the survey. Names were recorded only in cases where respondents explicitly consented, and such information was used solely for internal verification purposes.

4. **Academic Identification of Surveyors**

Surveyors introduced themselves as students associated with the Indian Statistical Institute and carried valid institutional identity cards during fieldwork in order to establish the academic nature of the study and build respondent confidence.

5. **Political Neutrality**

The survey intentionally avoided questions related to political preference, union affiliation, labour disputes, or ideological matters so that respondents would not feel pressured or uncomfortable during participation.

6. **Confidential Data Handling**

All collected information has been stored in password-protected institutional systems and is intended strictly for academic and research purposes. The data will not be shared publicly or used outside the scope of the study.

These practices were adopted to ensure that the survey process remained respectful, transparent, and appropriate for an academic field study involving informal-sector respondents.

Part II

Survey Design and Data

Chapter 4

Survey Design

4.1. Design Philosophy

This survey has been designed to collect primary data from street vendors selected through a systematic field sampling procedure across different market locations. The aim is to ensure a broad and spatially distributed coverage of vendors, so that differences in business practices, customer flow, and UPI usage can be studied across shops and road segments without concentrating on a single type of location. The survey seeks to understand both the economic and behavioural dimensions of UPI adoption among vendors. It also captures the main factors that encourage UPI adoption, including customer demand, convenience, competition, and related fee arrangements.

The design of the questionnaire was debated over several class sessions and iterated through a pilot phase. The questionnaire was built to answer three broad questions:

1. Who is the vendor?
2. How is UPI used?
3. What difference does UPI make?

It was designed to combine factual business data, adoption behavior, and personal perceptions in a way that is suitable for statistical analysis. The questionnaire was designed to capture three broad kinds of variables.

First, it includes **categorical background variables** such as gender, age group, type of goods sold, location, and mode of operation, which are used to describe the sample and compare vendors across groups.

Second, it collects **UPI adoption and business outcome variables**, including

whether UPI is accepted, share of UPI users, fee arrangements, number of customers served, cash-versus-UPI split, daily earnings, revenue variation, and customers lost due to change shortage; these address adoption intensity, sales volume, and possible business effects.

Third, it gathers **perceptual or opinion variables** on transaction speed, ease of tracking earnings, saving behavior, digital trust, and payment preference, which help capture how vendors experience UPI in practice.

A final **surveyor observation variable** section to improve consistency and to flag responses where necessary.

4.2. Questionnaire Structure

1. Vendor Profile

This section collects background information about the respondent and the business. It includes gender, age group, goods or services sold, type of goods, location, years in business, and mode of operation. The purpose of this section is to describe the sample and to create variables for comparison across groups.

2. UPI Usage and Adoption History

This section measures whether the vendor accepts UPI and how strongly they use it. It asks when and why UPI was adopted, what proportion of customers pay digitally, whether the vendor pays any UPI-related fees, and how often technical problems occur. It also helps capture the practical costs of digital payment use, such as device rent, transaction charges, and network failure, which may affect continued adoption.

3. Sales, Customer Flow, and Revenue Management

This section focuses on the vendor's business activity and revenue patterns. It includes average ticket size, total customers served per day, whether daily collections are steady or volatile, earnings on the busiest and slowest days in the last week, customers served in the last hour, and the share of those customers who paid by UPI. It also asks whether UPI makes earnings easier to track, whether it changes saving behavior, and how business performance has changed compared with the previous year. These questions are designed to measure customer volume, revenue variation, and possible financial benefits of UPI, including whether digital payments improve record keeping or reduce cash-handling problems.

4. Perceptions of Efficiency and Trust

This section captures the vendor's opinions about UPI and cash. It asks whether UPI has made transactions faster or slower, how often customers are lost because of lack

of change, whether digital payment records create concern, whether customers leave due to UPI delays, and whether the vendor prefers UPI, cash, or both. This section is important because adoption is not only a matter of access, but also of experience. It helps explain whether vendors see UPI as convenient, trustworthy, efficient, or burdensome.

5. Surveyor Internal Consistency Check

This final section is completed by the surveyor immediately after the interview. It checks whether the responses are logically consistent, for example, whether the reported UPI and cash shares add up reasonably, whether the one-hour UPI share is consistent with the daily UPI estimate, whether a QR code was visibly present, and whether the interview atmosphere seemed cooperative or suspicious. This section improves data quality by flagging possible errors that must be taken care of in pre-processing.

4.3. Sampling Strategy

Reconnaissance Survey

Before data collection, two teams of students walked across two different assigned segments of the B.T. Road corridor. Each vendor was recorded on a Mapping Sheet noting location, language preferred, category (Perishable/Non Perishable), and visible UPI status (presence of a QR code). This Recce served three purposes: (i) The approach ensured coverage of the study region while avoiding selection bias; (ii) provided a baseline estimate of local UPI adoption prevalence and rate of cooperation; and (iii) it familiarized surveyors with vendor locations, facilitating efficient deployment during the main survey phase.

Pilot Survey

A short pilot survey of 15 vendors was conducted in the Dunlop Market Area. The response from the shop owners was satisfactory and thus our questionnaire was finalized with one observation that a lot of shop owners were not comfortable with giving info about their daily earnings.

Final Survey

Systematic random sampling was used for vendor selection. For each road, two survey teams covered the two sides of the road separately. On each side, the first shop was selected randomly from among the first five shops, and then every fifth shop was interviewed thereafter. If a shop was closed, then the shop right before it was surveyed. Surveyors waited for the consent of the shop owner and then proceeded with the survey which lasted about 10 minutes on average for each.

The final questionnaire is available here :- [Statistics Comprehensive Class Project Questionnaire](#)

Chapter 5

Data Pre-Processing

5.1. Data Cleaning

1. During data cleaning, several typographical and coding inconsistencies were corrected. For example, entries in the location column such as “KFC” and “near KFC” were standardized to a single consistent label, such as “KFC”. A few implausible values were also identified and flagged.
2. The most important inconsistencies appeared in the last 10 columns. Since the combined count of UPI and cash users should ideally be close to 10, rows with totals that were unusually high or low, such as 6 or 14, were flagged as errors. A few rows also contained an excessive number of missing values.
3. Some basic transformations were carried out in the income-related variables. For example, ranges written as “₹x–₹y” were converted to “x–y”, entries such as “above x” were rewritten as “ $> x$,” and “below x” was either recorded as “x” or approximated as 75% of x , depending on the context. If during any further analysis a value was required instead of bins in the form of entries, only then was it converted into a value following the assumptions for the test. Also, the current business duration was approximated to nearest year.
4. Rows containing obvious errors were highlighted in red for easier identification during review. For missing values, only two entries were imputed in ticket size. In such cases, the mean of the entries that lie in the same shop categories was used, such as fruit shops. However, when multiple values were missing in a row, the entry was retained as NA rather than being imputed. At small scales, this seems plausible because similar ticket sizes are expected in same shop categories.
5. For certain questions, variables such as age were later encoded in a binary form which is appropriate, since splitting them into more sub-categories would result

in insufficient data within each group. In any further analysis, the models were developed after making the necessary assumptions and combining categories to ensure fair representation across both ends.

Categorical Encoding

We also converted categorical responses into numerical codes, such as assigning “*Yes* = 1” and “*No* = 0”. Complete categorical coding is provided in the [Appendix](#).

Part III

Descriptive and Exploratory Analysis

Chapter 6

Demographic Profile of Street Vendors

6.1. Gender Distribution in the Vending Sector

The gender composition of street vendors is a critical metric for understanding the socio-economic dynamics of the informal economy. Our survey of 129 respondents highlights a significant gender imbalance within the local street vending ecosystem.

Gender Distribution of Street Vendors

n = 129 respondents

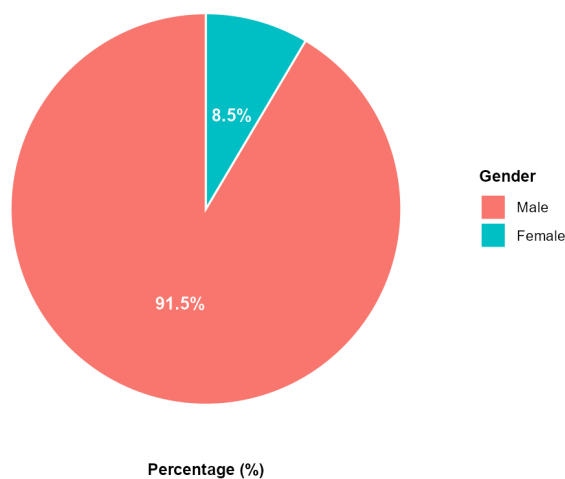


Figure 6.1. Gender Distribution (Pie Chart)

As shown in Figure 1.1, males constitute a staggering 91.5% of the workforce, while females account for only 8.5%.

6.2. Age Group Analysis and Workforce Maturity

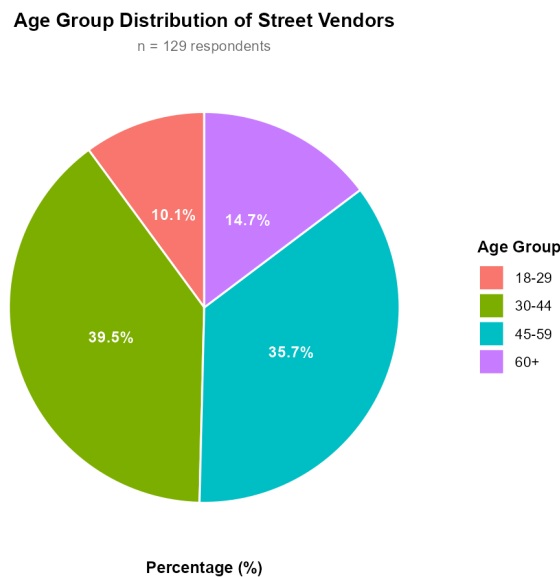


Figure 6.2. Age Group Distribution (Pie Chart)

The survey data regarding the age distribution of street vendors reveals a workforce heavily centered in middle-age brackets.

- **Dominant Demographic:** The workforce is primarily composed of individuals aged 30–59, who collectively represent **75.2%** of all respondents.
- **Primary Cohort:** The **30–44** age group is the single largest segment, accounting for **39.5%** of the total.
- **Youth Participation:** There is a notable lack of younger participants, with the **18–29** demographic being the smallest group at only **10.1%**.
- **Senior Presence:** Approximately **14.7%** of the vendors are aged 60 or older, indicating significant longevity or late-career entry in the sector.

6.3. Type of Goods

The survey data indicates a nearly balanced distribution of inventory types among the street vendors. **Perishable goods** (such as food and flowers) account for **51.2%** of the total, while **non-perishable goods** (such as clothing and electronics) comprise the remaining **48.8%**.

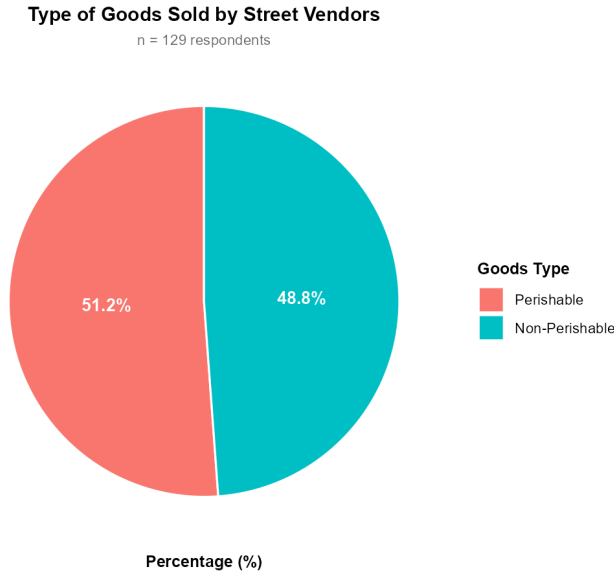


Figure 6.3. Type of Goods Sold by Street Vendors ($n = 129$)

6.4. Analysis of Market Longevity

The survey data for business duration (Figure 6.4) highlights a highly skewed market structure. The **0–5 years** category represents the highest density of vendors, accounting for nearly one-third of the total respondents.

As the years in business increase, a sharp decline is observed, typical of informal sectors with low barriers to entry but significant long-term survival challenges. However, the presence of vendors with over **50 years** of experience points to the role of street vending as a permanent, lifelong livelihood for a resilient minority of the population.

6.5. Operational Insights:

Analysis of the operational modes among 129 surveyed street vendors reveals a high degree of geographic stability. The vast majority of respondents, **89.9%**, utilize a **Permanent Stall** for their business activities. This is followed by a small minority of **Mobile Cart** users at **7%**, and **Pavement** vendors at **3.1%**. This distribution suggests that the local vending economy is largely composed of fixed-point retail rather than nomadic or highly mobile operations.

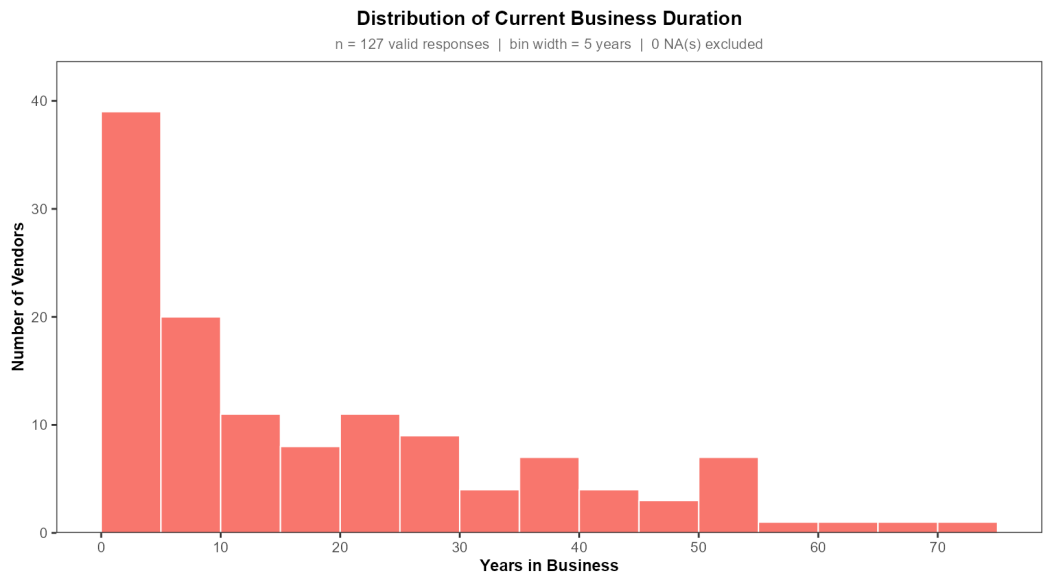


Figure 6.4. Distribution of Current Business Duration ($n = 127$)

6.6. Cross-Tabulation Analysis:

The grouped bar chart (Figure 6.6) illustrates the operational preferences of 118 male and 11 female vendors. A dominant trend is observed where **Permanent Stalls** serve as the primary mode of operation for both groups, accounting for 107 males and 9 females.

While male vendors show a slightly higher variety in operational modes in absolute terms (8 carts and 3 pavement setups), the relative scarcity of **Mobile Carts** and **Pavement** operations among females suggests that fixed-point vending is the most accessible or preferred business model for both genders in this specific market.

6.7. Generational Operational Trends:

The distribution across age groups (Figure 6.7) reaffirms the centrality of **Permanent Stalls** in the local street vending economy. Notably, the youngest demographic (18–29) is found exclusively in fixed stalls.

The 45–59 age cohort is the most operationally diverse, being the sole group to report **Pavement** vending. This suggests that while permanent infrastructure is the goal for most, mid-to-late career vendors are the most likely to employ alternative or low-infrastructure methods of operation.

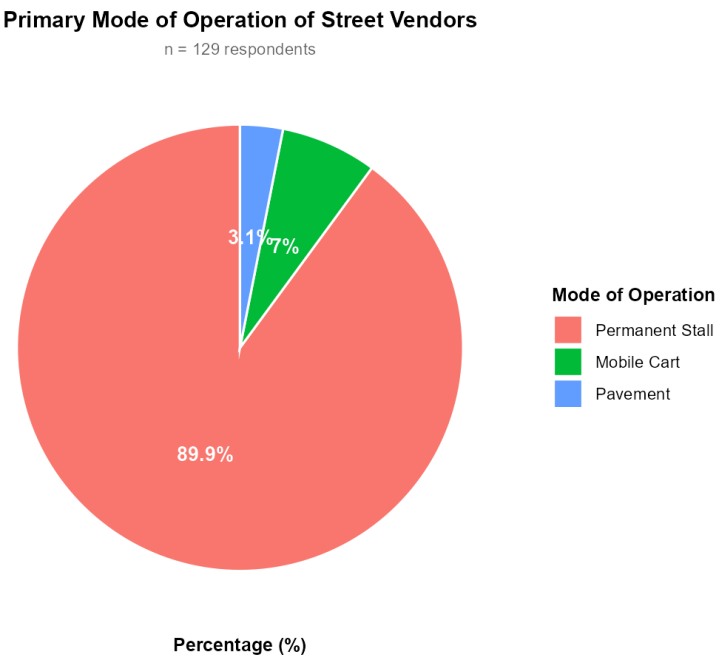


Figure 6.5. Primary Mode of Operation of Street Vendors ($n = 129$)

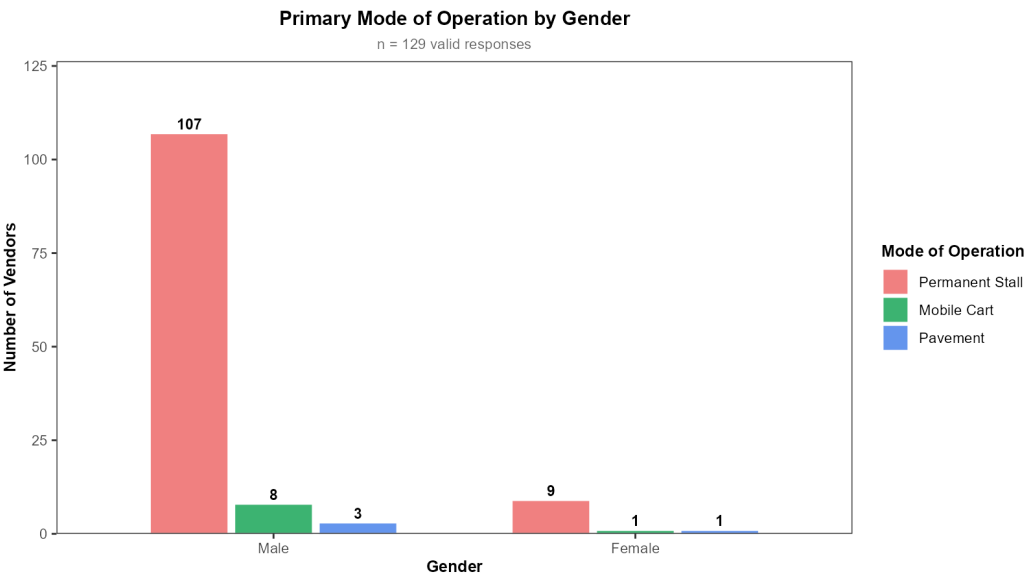


Figure 6.6. Gender vs. Primary Mode of Operation ($n = 129$)

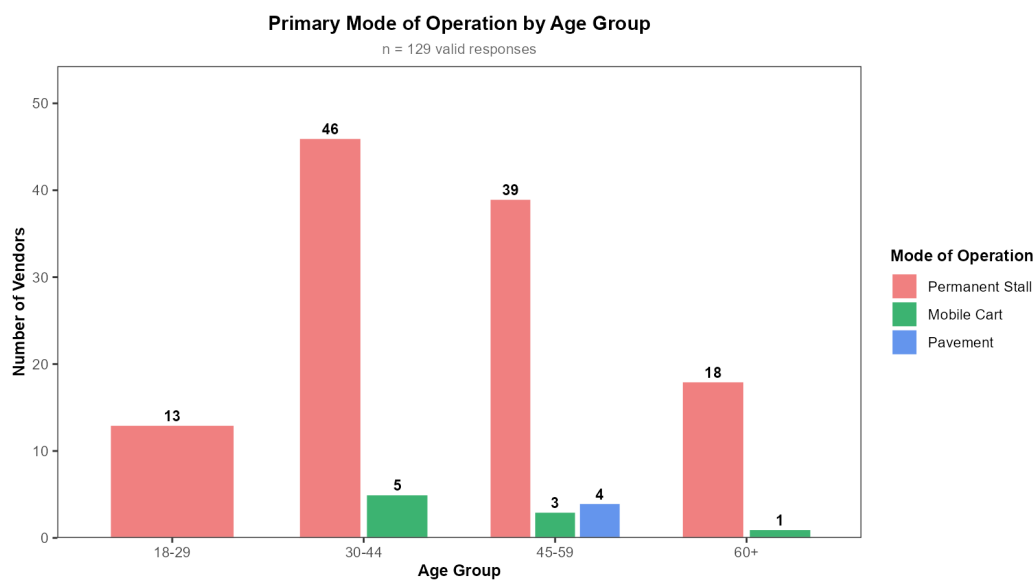


Figure 6.7. Age Group vs. Primary Mode of Operation ($n = 129$)

Chapter 7

The UPI Ecosystem

7.1. UPI Acceptance Among Street Vendors

The pie chart illustrates a near-universal adoption rate of digital payments within the street vending community.

- **High Adoption:** A staggering **93%** of the surveyed vendors have integrated UPI into their business operations.
- **Low Resistance:** Only **7%** of vendors remain cash-only, highlighting that digital payment infrastructure has become a standard requirement for micro-entrepreneurs.

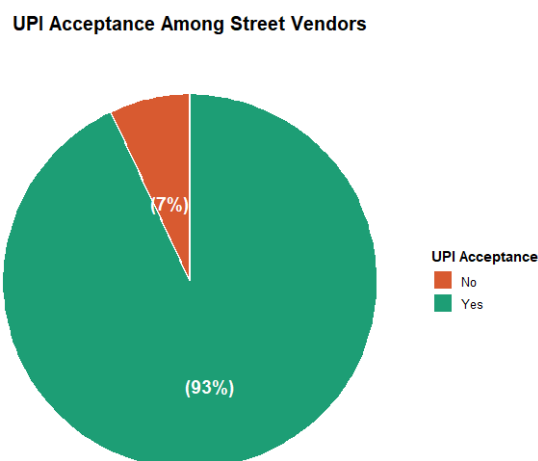


Figure 7.1. UPI Acceptance Percentage

7.2. UPI Adoption Reasons

This horizontal bar chart identifies the primary drivers behind the shift to digital payments, with a focus on market-driven necessity.

- **Customer Demand (70 vendors):** By far the most significant driver, indicating that vendors are adapting to the changing payment preferences of their clientele.
- **Ease & Convenience (32 vendors):** Highlighting the operational benefits of not handling physical change and direct bank settlements.
- **Social and Peer Influence (24 vendors):** Suggests a community-led transition where vendors learn from and mimic the success of their peers.
- **External Events (7 vendors):** A smaller portion attributed their shift to external factors.

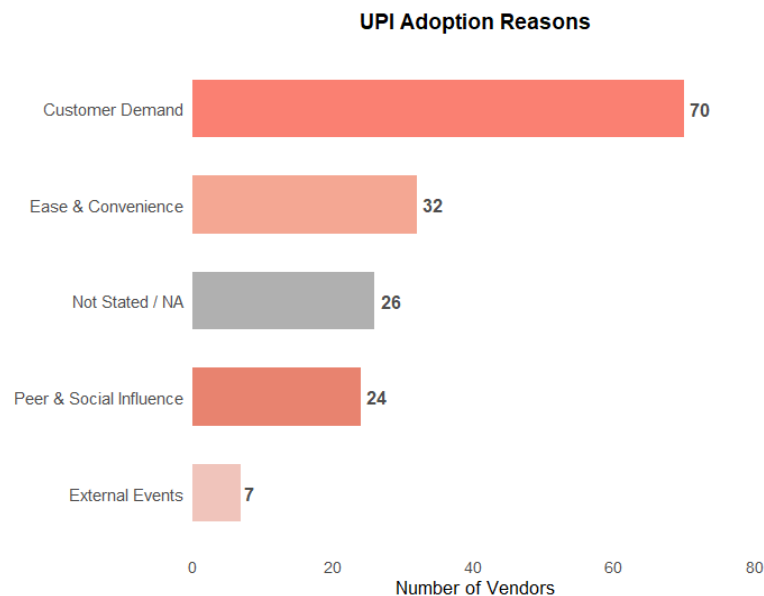


Figure 7.2. Primary Drivers for UPI Adoption

7.3. UPI Rent and Cost Structure

This plot examines the financial burden or service costs associated with using UPI platforms (likely referring to soundbox devices or merchant fees).

- **No Rent (63 vendors):** Nearly half of the respondents use UPI without incurring recurring costs, likely using personal QR codes.
- **Device-Based Rent (44 vendors):** A significant number pay for hardware, such as “Soundboxes,” which provide audio confirmation.

- **Per Sale Commission (13 vendors):** A smaller group operates on a transaction-fee model.

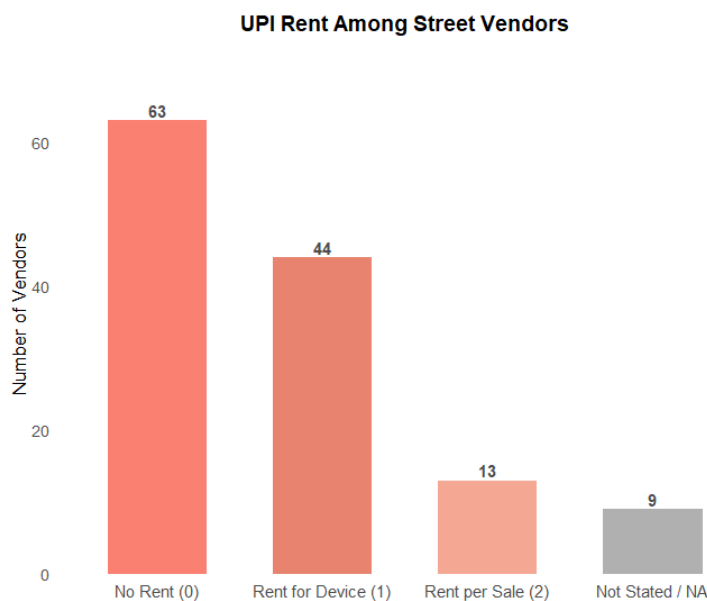


Figure 7.3. Cost Structure of UPI Usage

7.4. UPI Transaction Frequency per 10 Customers

This bar chart (figure 7.4) gives a rough idea about the “digital intensity” of transactions.

- **Most Common Band (6–8 range):** The majority of vendors (**54**) report that between 6 to 8 out of every 10 customers pay via UPI. This suggests that for a large section of vendors, UPI has already become part of routine business activity rather than an occasional alternative to cash.
- **The Upper End (9–10 range):** **22** vendors report that UPI dominates their transactions, with almost all transactions (90 to 100%) being digital.
- **The Lower End (0–5 range):** A combined **45** vendors see digital usage from half or fewer of their customers.

7.5. Network Issue Frequency Among Street Vendors

Figure 7.5 assesses the technical reliability of the UPI system in a street-level environment, by looking at how frequently they experience such disruptions during trade.

- **High Reliability (Rarely):** A sizable faction of **82** vendors reported that they

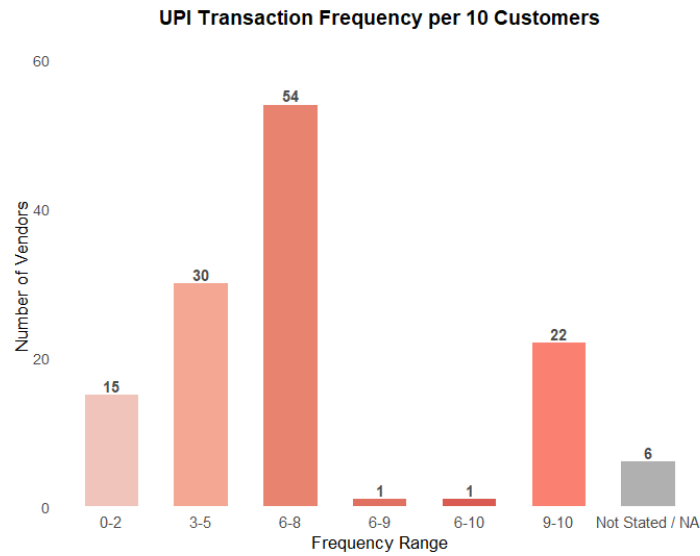


Figure 7.4. UPI Transaction Density per 10 Customers

rarely face network issues.

- **Manageable Disruptions:** Only a small minority face **Daily (7)** or **Weekly (17)** issues.
- Another **14** vendors reported occasional **monthly** disruptions.

Reliability plays a major role in whether vendors continue using digital payments in high-frequency retail settings.

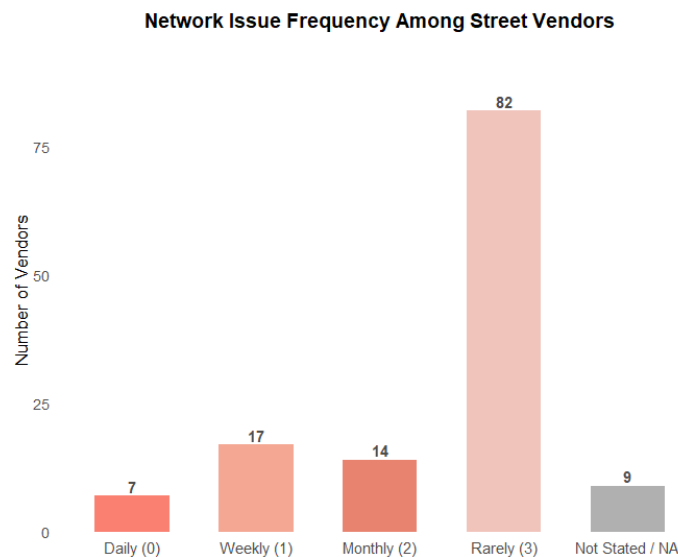


Figure 7.5. Frequency of Technical and Network Disruptions

Chapter 8

Analysis of Business Earnings and Payment Trends

8.1. Analysis of Ticket Size Distribution

Pie Chart showing Distribution of Ticket Sizes

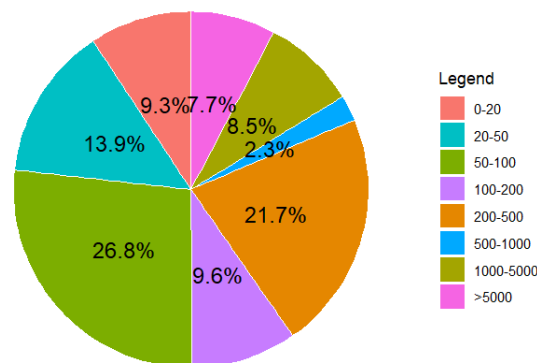


Figure 8.1. Distribution of Ticket Sizes

This pie chart illustrates the breakdown of ticket sizes by percentage. It helps us understand the most common transaction ranges within the data.

- **Dominant Range:** The **50–100** category is the largest segment, making up **26.8%** of all tickets. This indicates that roughly one-fourth of the transactions fall into this mid-low price bracket.
- **Secondary High-Volume Range:** The **200–500** category is the second most frequent, accounting for **21.7%** of the total.

- **Lower Value Tickets:** Smaller tickets (under 50) combine for about **23.2%** (9.3% for 0–20 and 13.9% for 20–50), showing a healthy volume of low-cost transactions.
- **High-Value Rarity:** Tickets over **500** represent the smallest portions of the chart. Specifically, the **500–1000** range is the least common at only **2.3%**. Interestingly, the largest tickets (**>5000**) actually occur more frequently (**7.7%**) than the mid-high range.

Summary: The majority of the business is driven by tickets between **50 and 500**. While very high-value tickets (**>5000**) are not the majority, they still hold a larger share than mid-range high-value tickets (500–1000).

8.2. Analysis of Earnings by Business Trajectory

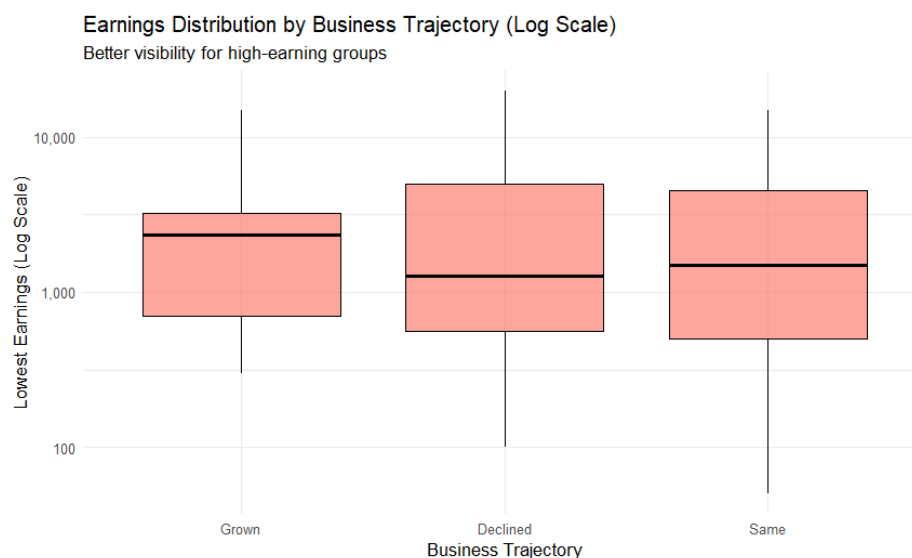


Figure 8.2. Earnings Distribution by Business Trajectory (Log Scale)

This chart compares the lowest earnings of businesses based on whether they have **Grown**, **Declined**, or stayed the **Same**. A log scale is used to better visualize the wide range of income levels.

- **Higher Average Earnings:** Businesses in the “**Grown**” category have the highest median earnings (the middle line in the box). This suggests that growth is linked to a higher baseline of financial success.
- **Greater Stability:** The “**Grown**” group has a smaller box and shorter “whiskers.” This means their earnings are more consistent and less likely to drop to extremely low levels compared to the other groups.
- **Higher Risk in Declining Businesses:** The “**Declined**” and “**Same**” groups show much more variation. Their long bottom whiskers indicate that these businesses are at a much higher risk of seeing earnings fall toward very low values (near

100).

- **Top Performers:** While the averages differ, the top whiskers for all three groups reach similar high levels (above 10,000). This shows that high-earning potential exists across all trajectories, though it is most consistent in growing businesses.

Summary: Growing businesses generally provide higher and more stable earnings, while declining or stagnant businesses face much higher unpredictability and financial risk.

8.3. Analysis of Earnings Distribution (Histogram)

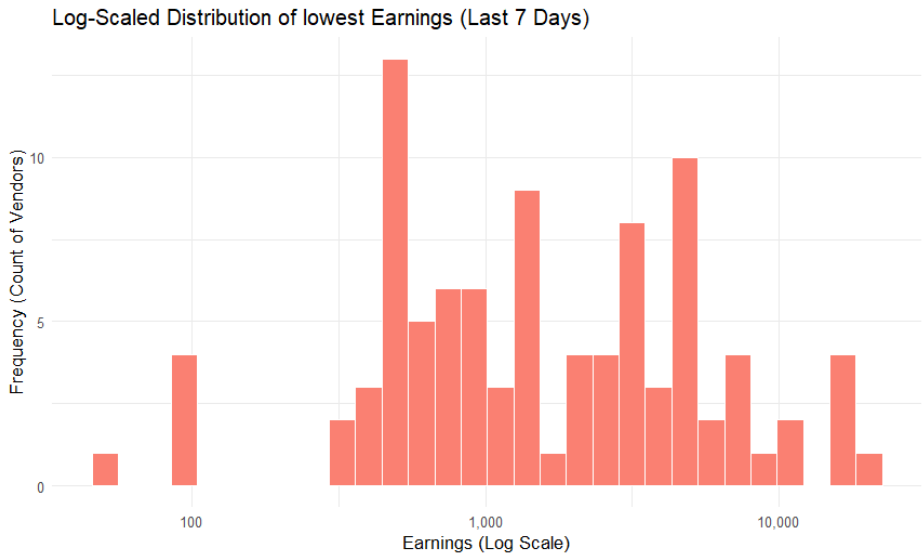


Figure 8.3. Log-Scaled Distribution of Lowest Earnings (Last 7 Days)

This histogram displays the distribution of lowest earnings for vendors over the last 7 days. By using a **log scale** on the x-axis, we can see the full range of vendors, from those earning very little to those earning over 10,000.

- **Most Common Earnings:** There is a significant peak (the tallest bar) just below 1,000. This suggests that a large portion of vendors falls into this specific earnings bracket.
- **Multi-Modal Distribution:** The graph shows several “peaks” (e.g., near 500, 1,500, and 5,000). This indicates that vendors tend to cluster into different performance tiers rather than following a single smooth curve.
- **Low-Earning Outliers:** A small number of vendors are visible on the far left, earning around 100 or less. These represent a minority that is significantly underperforming compared to the rest of the group.
- **High-End Spread:** On the right side of the graph, the bars are more spread out as earnings exceed 10,000. While there are fewer vendors at this level, the log scale reveals that high-earning potential is quite broad.

Summary: The majority of vendors earn between **500 and 5,000**. The presence of multiple peaks suggests that different business strategies or categories may be driving distinct levels of success within the vendor population.

8.4. Analysis of Highest Earnings by Business Trajectory

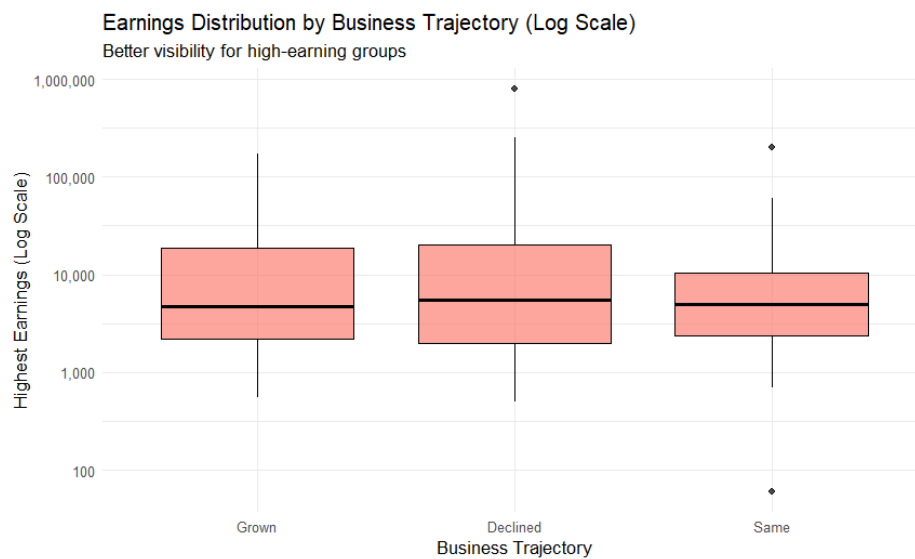


Figure 8.4. Highest Earnings Distribution by Business Trajectory (Log Scale)

This boxplot examines the **highest earnings** across different business trajectories. Since the y-axis is on a **log scale**, small visual gaps actually represent large differences in dollar value.

- **Consistent Median:** The median earnings (the middle black lines) are remarkably similar across all three groups—**Grown**, **Declined**, and **Same**. All three sit slightly above the **5,000** mark, suggesting that a business’s current trajectory doesn’t necessarily dictate its peak earning potential.
- **Outliers in Declining Businesses:** The “**Declined**” group shows the highest individual outlier, reaching near **1,000,000**. This indicates that even though these businesses are shrinking, some individuals within this category still achieve exceptionally high peak earnings.
- **Wider Range in “Declined” and “Grown”:** These two groups have larger boxes (Interquartile Range) compared to the “**Same**” group. This shows that businesses that are actively changing (either growing or shrinking) have more unpredictable peak earnings than those that stay the same.
- **Downside Performance:** The “**Same**” category includes a very low outlier (near **100**). This suggests that while stagnant businesses are generally stable, there is still a risk of very low peak performance in some cases.

Summary: While median peak earnings are stable across all categories, businesses that are **declining** or **growing** show much higher variance and higher potential for extreme “top-tier” earnings compared to those remaining the same.

8.5. Analysis of Highest Earnings Distribution (Histogram)

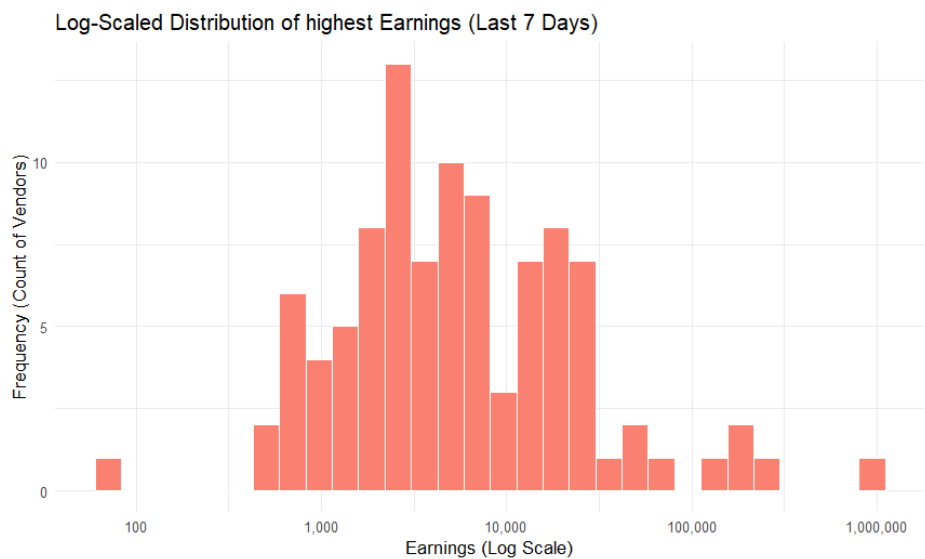


Figure 8.5. Log-Scaled Distribution of Highest Earnings (Last 7 Days)

This histogram shows the distribution of the **highest earnings** achieved by vendors over the last 7 days. The x-axis uses a **log scale** to visualize everything from small-scale earnings to values reaching 1,000,000.

- **Primary Earning Peak:** The highest concentration of vendors (the tallest bar) occurs between **2,000 and 3,000**. This is the most common "peak" earning level for the majority of the population.
- **Secondary High-Earning Tiers:** There are visible clusters around **5,000–8,000** and again near **20,000**. This indicates that there are distinct groups of high-performing vendors who consistently hit higher earning brackets.
- **Significant Outliers:** The log scale reveals rare but extreme success stories on the far right, with some vendors' highest earnings reaching between **100,000 and 1,000,000**.
- **Overall Range:** Most vendors see their highest weekly earnings fall between **1,000 and 30,000**. Very few vendors have a "highest earning" day that remains below 500.

Summary: While the typical "top" earning day for a vendor is in the low thousands, the distribution has a **long right tail**, showing that the platform supports a small number of extremely high-volume earners.

8.6. Analysis of UPI Paying Customers (Last Hour)

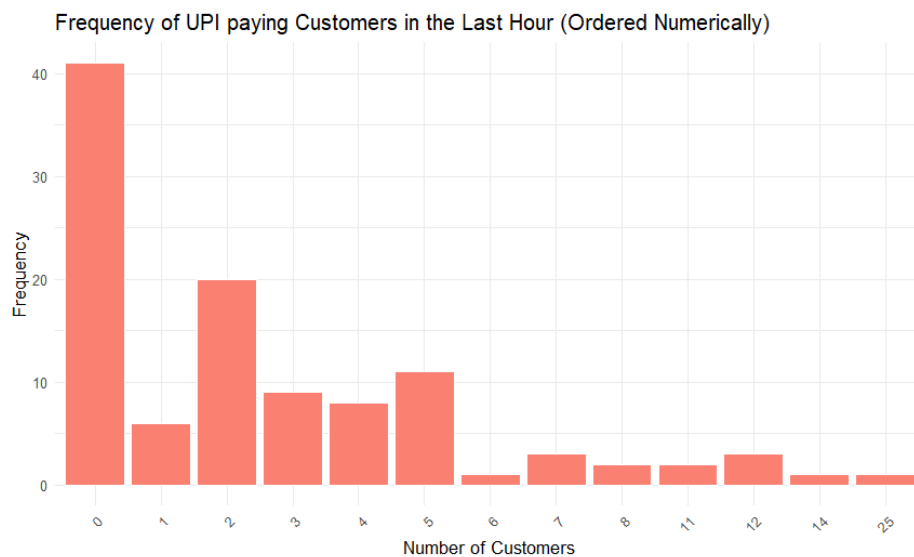


Figure 8.6. Frequency of UPI Paying Customers in the Last Hour

This bar chart shows how many UPI customers were served by vendors in the last hour.

- **Most Common Scenario:** The tallest bar is at **0 customers**, with a frequency of over 40. This indicates that in any given hour, a large number of vendors do not process any UPI transactions.
- **Active Range:** For vendors who did have customers, the most common numbers were **2 customers** and **5 customers**.
- **Rapid Drop-off:** There is a significant decrease in frequency as the number of customers increases. Serving more than 5 UPI customers in a single hour is relatively rare.
- **High-Volume Outliers:** There are very few instances where vendors served a high volume of customers (such as **12, 14, or 25**) in one hour. These likely represent high-traffic locations or peak rush periods.

Summary: UPI usage is currently concentrated among a smaller group of active vendors, with most vendors seeing low to zero UPI activity per hour. The data shows a **right-skewed distribution**, meaning high customer volume is the exception rather than the rule.

8.7. Analysis of Total Customers (Last Hour)

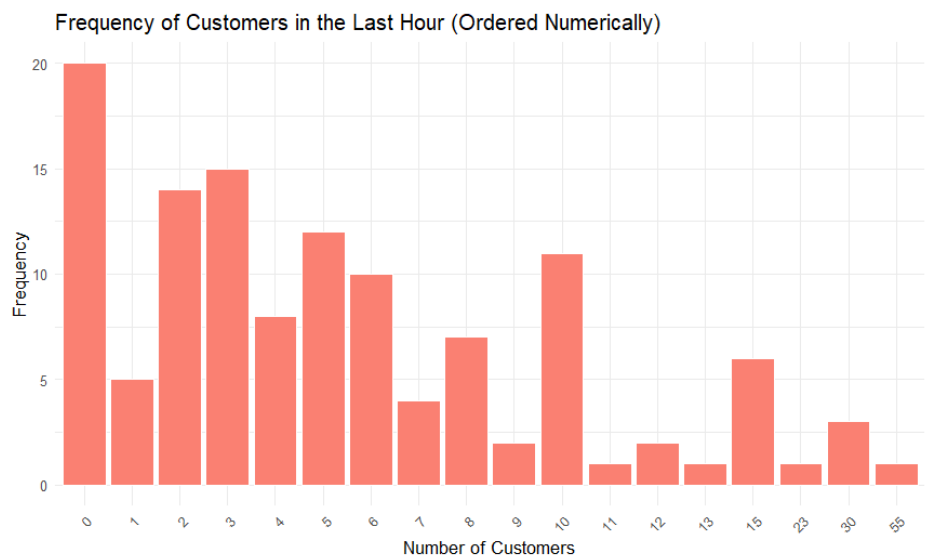


Figure 8.7. Frequency of Total Customers in the Last Hour

This bar chart shows the total number of customers (using all payment methods) served by vendors in the last hour.

- **Peak Frequency at Zero:** Similar to UPI transactions, the most frequent observation is **0 customers**, occurring 20 times. This shows that at any given hour, a significant portion of vendors are not seeing foot traffic.
- **Core Customer Range:** For active vendors, the most common customer counts fall between **2 and 6** per hour. There is also a notable secondary peak at **10 customers**, representing a "busy" tier of vendors.
- **Higher Overall Traffic:** Compared to the UPI-only chart, there is a much higher frequency of vendors serving more than 10 customers (with peaks at 15, 23, 30, and even 55). This suggests that total foot traffic is significantly higher than UPI-specific traffic.
- **Extreme Outliers:** The chart shows a very long "tail," with some vendors serving as many as **30 to 55 customers** in a single hour. These are likely high-volume hubs or businesses experiencing a peak rush.

Summary: While many vendors experience quiet hours, the total customer distribution is much broader than UPI usage alone. This indicates that while foot traffic can be very high for some, a large portion of those customers may still be using other payment methods like cash.

8.8. Analysis of Average Customers per Day

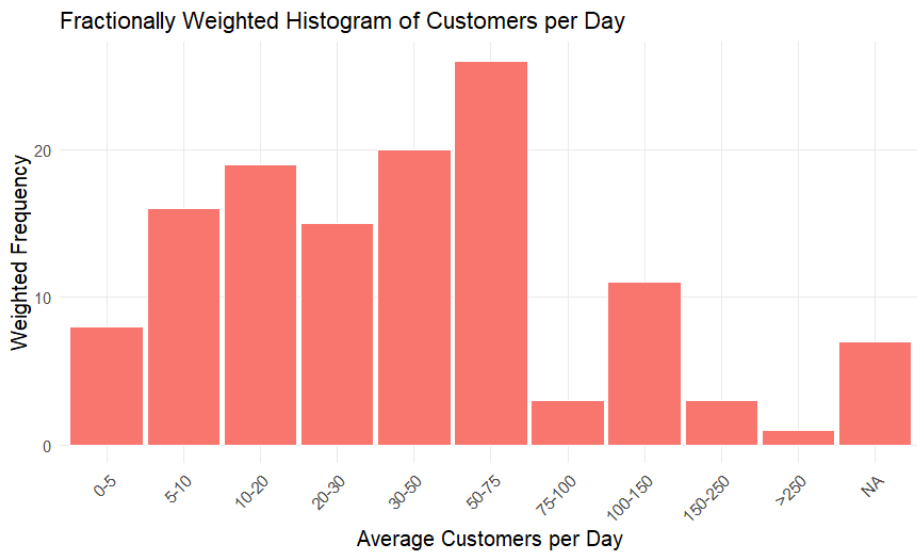


Figure 8.8. Fractionally Weighted Histogram of Customers per Day

This fractionally weighted histogram shows the distribution of average daily customers across the vendor population. Weighting the data ensures that the frequencies accurately reflect the proportions within the dataset.

- **Primary Peak:** The most common average is **50–75 customers per day**, which represents the highest weighted frequency. This suggests a significant "middle class" of vendors with steady daily foot traffic.
- **High-Volume Clusters:** There is a broad concentration of vendors serving between **10 and 75 customers daily**. This indicates that the majority of vendors on the platform handle a moderate to high volume of daily transactions.
- **Secondary Surge:** A smaller but notable peak appears in the **100–150** range. This highlights a subset of high-performance vendors who serve a much larger customer base than the average.
- **Extreme Highs and Missing Data:** A very small fraction of vendors serves **over 250 customers per day**. Additionally, there is a small "NA" category, representing vendors where customer count data was not available or recorded.

Summary: The vendor base is healthy, with the majority averaging over 30 customers per day. The data shows a **multi-modal distribution**, where most vendors cluster around the 50–75 range, but a distinct group of high-capacity vendors operates at nearly double that volume.

8.9. Analysis of Perception: Does UPI Help Save More?

Pie Chart showing perception of people if UPI helps save more

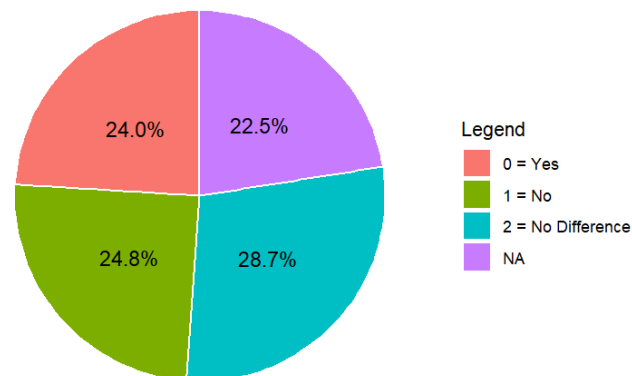


Figure 8.9. Perception of People if UPI Helps Save More

This pie chart shows how people perceive the impact of using UPI on their ability to save money. The data is divided almost equally, showing a significant split in public opinion.

- **Mixed Opinions:** The responses are nearly evenly split between **"Yes"** (24%), **"No"** (24.8%), and **"No Difference"** (28.7%). This suggests there is no clear consensus on whether UPI encourages or discourages saving.
- **Neutrality as the Leading View:** The largest group, at **28.7%**, believes that using UPI makes **"No Difference"** to their savings. This group likely views UPI strictly as a convenient payment tool rather than a financial management influence.
- **The "Saving vs. Spending" Debate:** Roughly **24%** feel UPI helps them save more (perhaps through better digital tracking), while a nearly identical **24.8%** feel it does not (perhaps due to the ease of impulsive digital spending).
- **Significant Missing Data:** A notable **22.5%** of respondents fall under the **"NA"** category. This represents a large portion of the population that either does not use UPI or did not provide a definitive answer.

Summary: Public perception regarding UPI and savings is highly fragmented. While nearly 30% see it as neutral, the rest of the population is almost perfectly divided on whether digital payments help or hinder their financial goals.

8.10. Analysis of Payment Method Distribution

Pie Chart showing distribution of Payment Methods

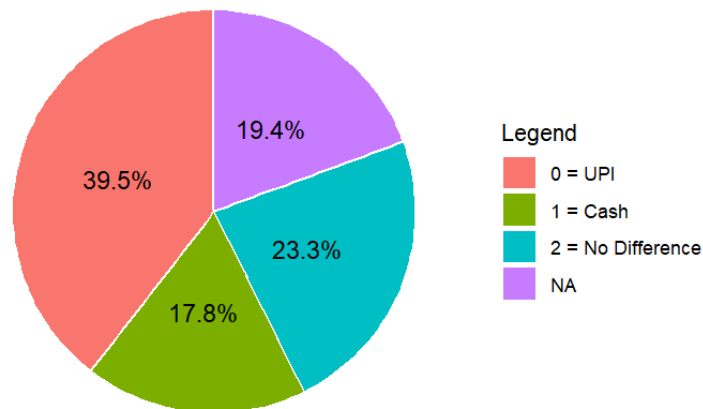


Figure 8.10. Perception of People if UPI helps them track their finances better

This pie chart shows how people perceive the impact of UPI on their capability to track their finances.

- **UPI Dominance:** Around **39.5%** of the total sample believe that **UPI** clearly helps them to track finances better than cash. This highlights a strong preference for digital, instant payments.
- **Cash Usage:** Approximately **17.8%** of the vendors prefer **Cash** accounts to track their finances. While it is still significant, it is preferred less than half as often as UPI.
- **Indifference:** A notable **23.3%** of respondents fall under **"No Difference."** This suggests that nearly a quarter of the group is comfortable using either cash or digital payments and does not have a strict preference.
- **Missing Data:** The **"NA"** category represents **19.4%**, which likely accounts for respondents who do not use these methods or did not answer the question.

Summary: The data shows a clear shift toward digital finance, with **UPI** being the preferred choice to track their finances for a large segment of the population. However, the high percentage of people in the "No Difference" category suggests that the transition from cash to digital is still in a flexible stage for many.

Chapter 9

Cash, UPI, and Behavioural Dynamics Among Street Vendors

9.1. Analysis of Preferred Mode of Transaction

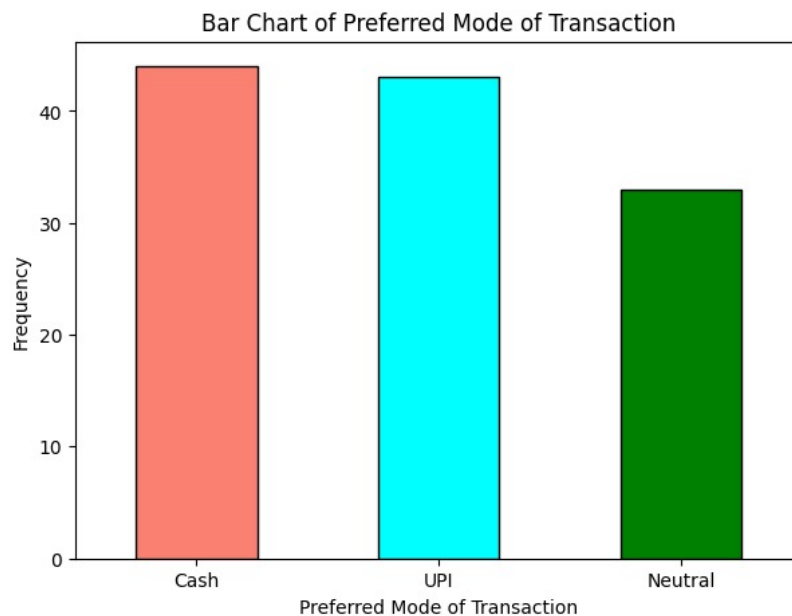


Figure 9.1. Bar Chart of Preferred Mode of Transaction

This bar chart shows which payment methods people prefer. By looking at the frequency (the number of people), we can see how popular each option is.

Key Findings:

- **Most Popular Choice:** Cash is the top preference, chosen by about 44 people. This shows that many people still feel most comfortable using physical money.
- **Digital Growth:** UPI is almost as popular as cash, with 43 people preferring it. This tiny difference shows that digital payments are now just as common as using cash.
- **Neutral Group:** About 33 people are "Neutral." This means a good portion of the group doesn't mind which method they use or they use both equally.

Summary: The data shows a very close race between Cash and UPI. While cash is technically the "winner," the results suggest that the community is split right down the middle between traditional money and digital apps.

Total Respondents: Approximately 120 people.

9.2. Analysis of Customers Lost Due to Lack of Cash

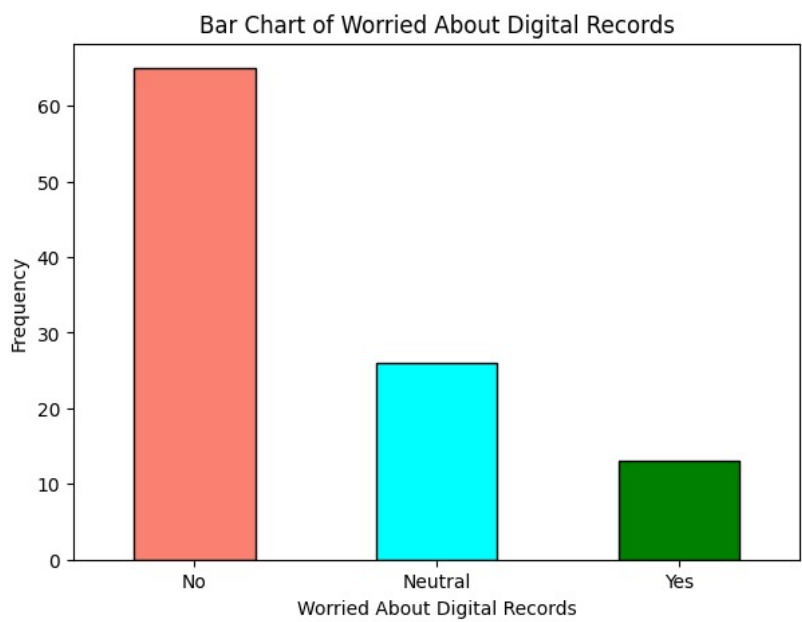


Figure 9.2. Bar Chart of Customer lost for not having cash

This chart explores how often businesses lose customers because they do not have physical cash available. It measures the number of customers lost (0, 10, or 20) against how frequently that occurs.

Key Findings:

- **Most Common Scenarios:** The bars for 0 and 10 customers lost are equal in height, both with a frequency of 2. This shows that in this sample, it is just as common to lose no customers as it is to lose 10 customers due to a lack of cash.

- **Extreme Cases:** Losing 20 customers is less common, with a frequency of only 1.
- **Data Trend:** While the sample size for this specific chart is small (total frequency of 5), it indicates that not having cash does lead to a measurable loss of business in most of the observed cases.

Summary: The data suggests that while some businesses stay unaffected (0 lost), a significant portion of the observations shows a loss of 10 to 20 customers. This highlights that even in a digital age, lacking cash can still result in missed sales opportunities.

Total Observations: 5 instances.

9.3. Analysis of Concern Regarding Digital Records

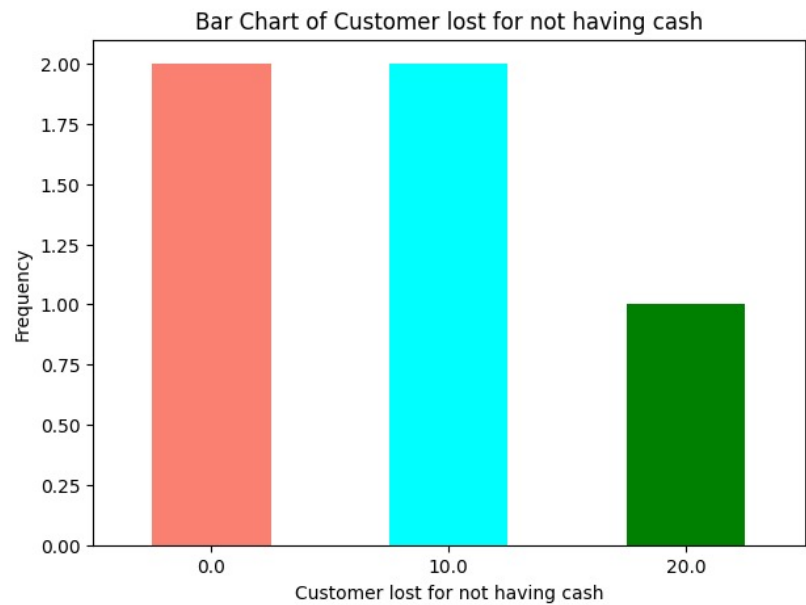


Figure 9.3. Bar Chart of Worried About Digital Records

This bar chart shows whether respondents are worried about their digital transaction records. It helps us understand the level of trust people have in digital payment history and data privacy.

Key Findings:

- **High Level of Trust:** The majority of respondents (65) answered "No," indicating they are not worried about digital records. This is the tallest bar by a significant margin, showing strong confidence in digital record-keeping.
- **Moderate Uncertainty:** About 26 people are "Neutral." These individuals may have slight doubts or simply do not give much thought to how their digital data is stored.

- **Low Level of Concern:** Only a small group of about 13 people answered "Yes." This shows that only a minority of the sample feels anxious or worried about their digital footprint.

Summary: The data clearly shows that most people are comfortable with digital records. The number of people who are not worried is more than double the number of neutral and concerned people combined. This suggests that privacy concerns are not a major barrier for this group when using digital payment methods.

Total Respondents: Approximately 104 people.

9.4. Analysis of Cash Usage Among the Last 10 Customers

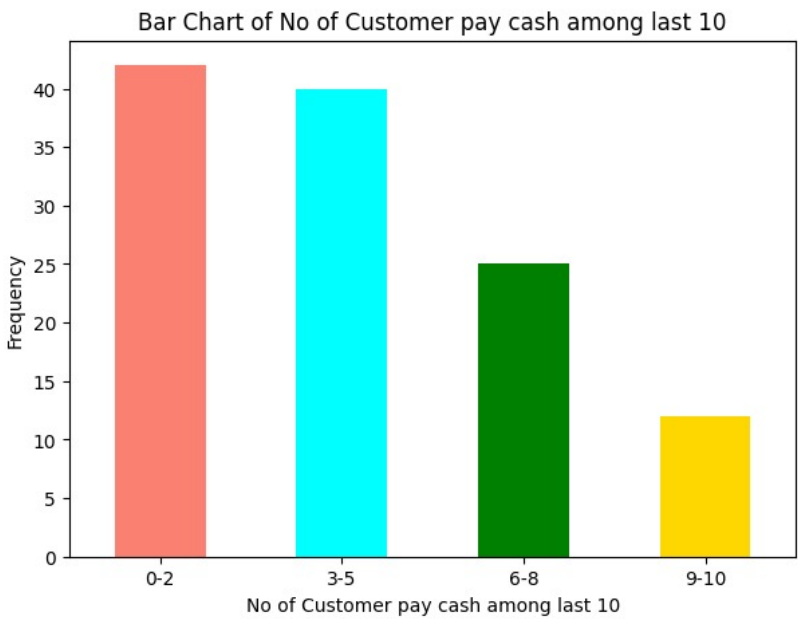


Figure 9.4. Bar Chart of No of Customer pay cash among last 10

This bar chart shows how many customers out of the last ten used cash for their transactions. It helps us understand the current density of cash payments in recent business activity.

Key Findings:

- **Low Cash Usage is Most Common:** The highest frequency (42) occurs in the 0-2 range. This means that in most observed cases, only a very small number of customers (2 or fewer out of 10) chose to pay with cash.
- **Moderate Cash Usage:** The 3-5 range also shows a high frequency of 40. Combined with the first group, this indicates that in the vast majority of instances, 50% or fewer customers are using cash.
- **High Cash Usage is Rare:** The frequency drops significantly for the 6-8 range (25) and the 9-10 range (12). It is much less common to see a group where almost everyone pays with cash.

Summary: The data shows a clear trend: digital or non-cash payments are likely more frequent. In more than 80% of the observations, 5 or fewer customers out of 10 used cash. This suggests that while cash is still used, it is no longer the majority choice for most customer groups in this sample.

Total Observations: Approximately 119 groups of 10 customers.

9.5. Analysis of Transaction Speed Using UPI

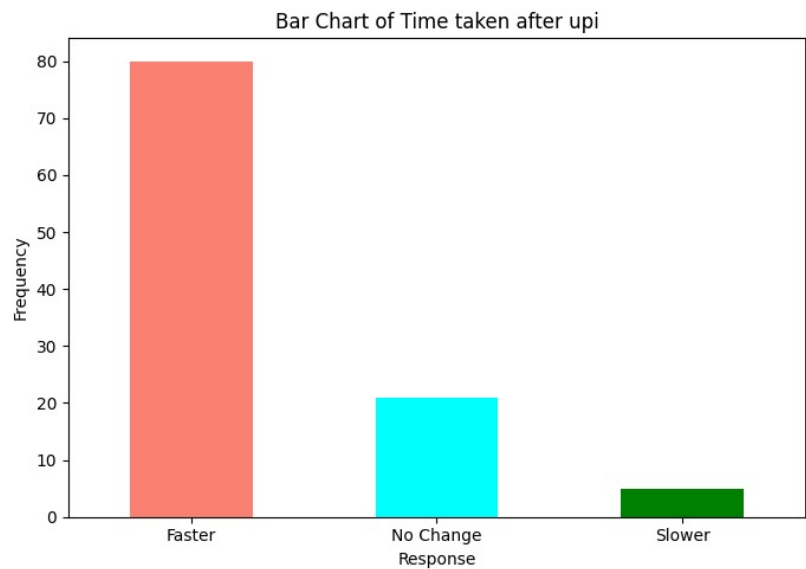


Figure 9.5. Bar Chart of Time taken after upi

This bar chart measures how the use of UPI has affected the time taken to complete a transaction. It categorizes the impact as "Faster," "No Change," or "Slower."

Key Findings:

- **Significant Efficiency Gain:** The overwhelming majority of respondents (80) reported that transactions are "Faster" with UPI. This is the dominant mode in the data, showing that speed is the primary benefit of digital payments.
- **Neutral Impact:** A smaller group of about 21 people felt there was "No Change" in the time taken compared to previous methods.
- **Minimal Delays:** Only a very small minority (5 people) found the process to be "Slower." These rare cases could be due to technical issues like poor network connectivity or server downtime.

Summary: The data provides strong evidence that UPI has improved transaction efficiency. Approximately 75% of the total sample experiences faster payments, confirming that digital transactions are successfully reducing wait times for the majority of users.

Total Respondents: 106 people.

9.6. Analysis of Customers Lost Due to Lack of UPI

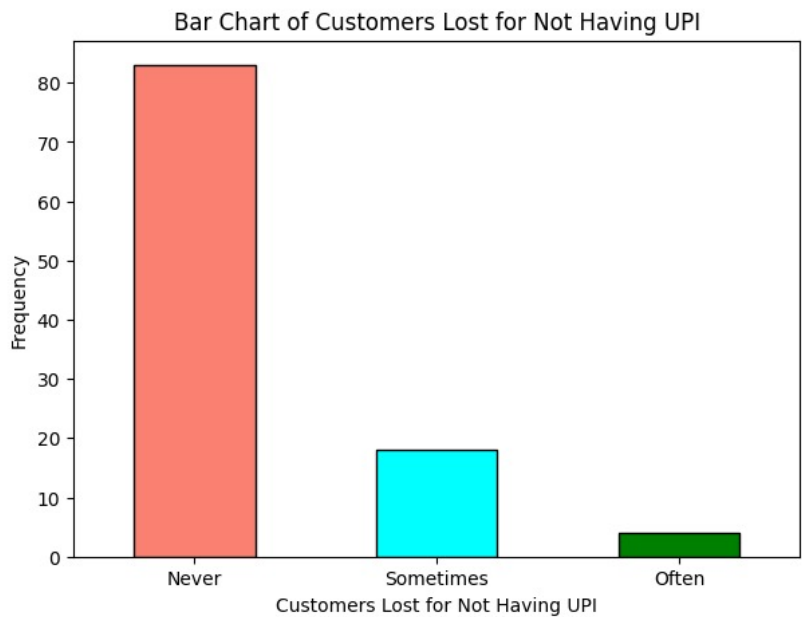


Figure 9.6. Bar Chart of Customers Lost for Not Having UPI

This bar chart examines how often businesses lose potential customers because they do not offer UPI as a payment option.

Key Findings:

- **High Availability/Low Loss:** The overwhelming majority of respondents (83) reported "Never" losing a customer due to a lack of UPI. This suggests that either most businesses already have UPI integrated or their specific customer base does not yet demand it as a deal-breaker.
- **Occasional Loss:** About 18 people mentioned they "Sometimes" lose customers. This indicates a small but present risk of losing sales for businesses that rely purely on traditional payment methods.
- **Frequent Loss:** Only a very small number of respondents (4) answered "Often." For this tiny minority, the absence of digital payment options is a significant obstacle to completing sales.

Summary: The data shows that for most people in this sample, not having UPI is not a major problem for business retention yet. However, since roughly 20% of the total respondents experience customer loss ("Sometimes" or "Often"), there is a clear incentive for businesses to adopt digital payments to avoid missing out on these transactions.

Chapter 10

Data at a Glance

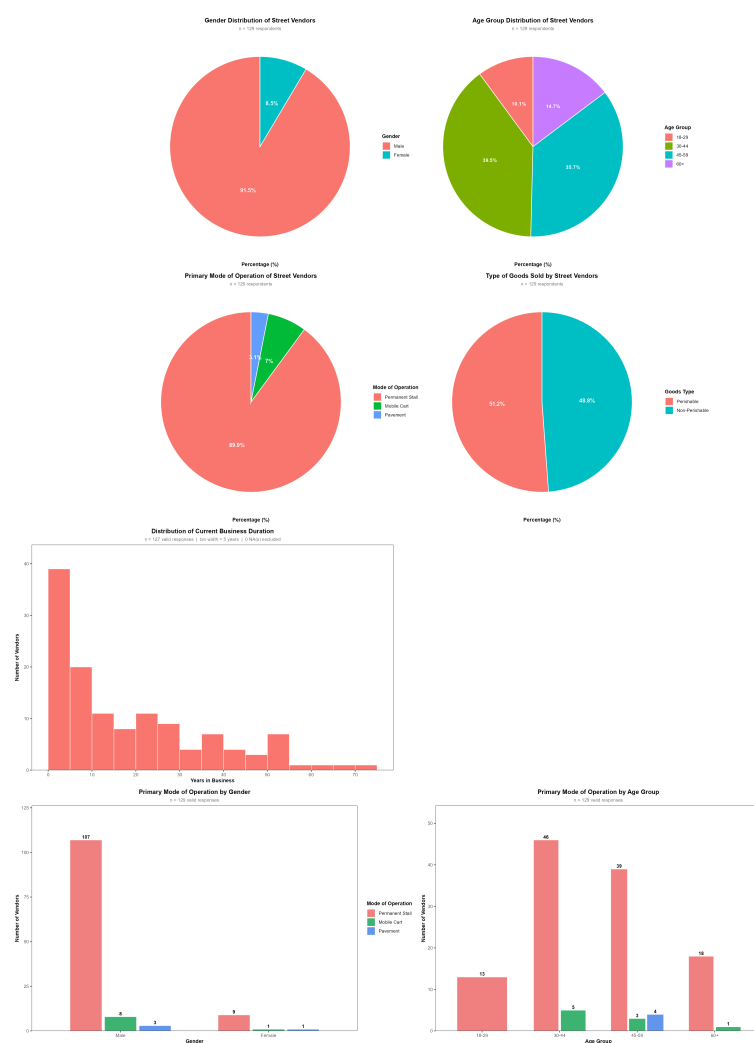


Figure 10.1. Question 1 Master Plot

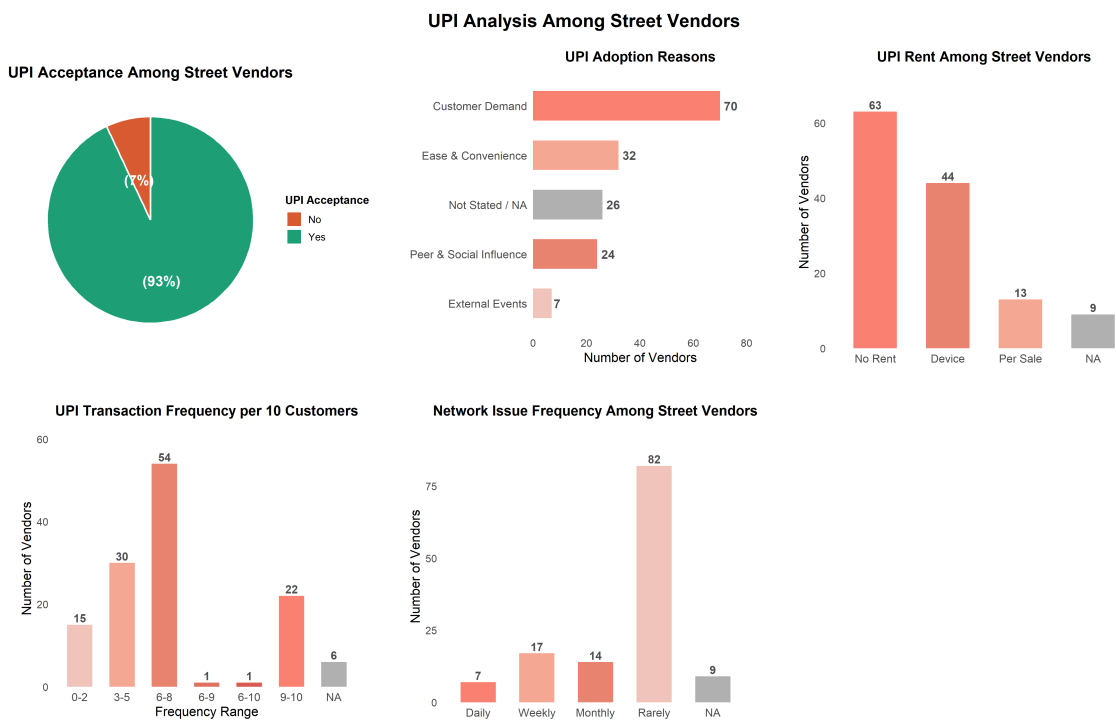


Figure 10.2. Question 2 Master Plot



Figure 10.3. Question 3 Master Plot

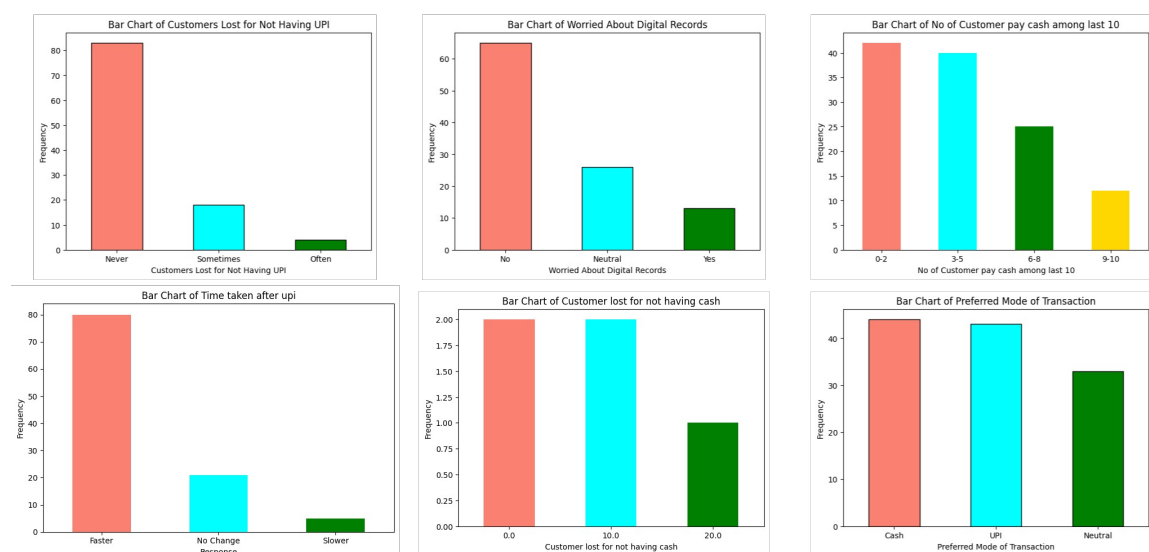


Figure 10.4. Question 4 Master Plot

Part IV

Inferential Analysis

Chapter 11

UPI Intensity Index

11.1. Reliability-Weighted Bayesian Estimation of UPI Intensity

A central objective of this study is the construction of a statistically stable and operationally meaningful measure of UPI adoption intensity at the vendor level. A naive empirical estimator built solely on short-term observations is inadequate here: customer arrivals exhibit substantial stochastic variability, and hour-long observation windows are too narrow to yield stable proportion estimates. On the other hand, long-run recall-based responses are susceptible to heaping effects, recall bias, and social-desirability distortion. The survey instrument was therefore designed to elicit two complementary sources of information:

1. a coarse ordinal estimate of long-run UPI usage frequency, and
2. an observational count of customers and UPI transactions in the preceding hour.

Rather than privileging one source over the other, we combine both within a reliability-weighted Bayesian framework.

11.1.1 Observed Data Structure

For vendor i , let n_i denote the total number of customers observed in the last hour and Y_i the number who paid via UPI. The survey also records a coarse long-run usage estimate through Question 12, mapped to a numerical midpoint m_i as follows:

Response Category	Midpoint m_i
0–2	0.10
3–5	0.40
6–8	0.70
9–10	0.95

The midpoint m_i serves as a noisy prior estimate of the latent probability that a randomly arriving customer pays via UPI.

11.1.2 Model Specification

Let p_i denote the true UPI usage probability for vendor i . Conditional on p_i , the observed transaction count follows a Binomial model, treating each customer arrival as an independent Bernoulli trial:

$$Y_i | p_i \sim \text{Binomial}(n_i, p_i).$$

The recall response is incorporated through a Beta prior conjugate to this likelihood:

$$p_i \sim \text{Beta}(a_i, b_i),$$

with the prior mean anchored at the recall midpoint,

$$\frac{a_i}{a_i + b_i} = m_i.$$

Writing $s_i = a_i + b_i$ for the *prior reliability weight* (effective prior sample size), the hyperparameters take the form

$$a_i = s_i m_i, \quad b_i = s_i (1 - m_i),$$

so that $p_i \sim \text{Beta}(s_i m_i, s_i (1 - m_i))$. A larger s_i reflects stronger confidence in the recall estimate; a smaller s_i allows the observed data to dominate.

11.1.3 Reliability Weighting via Internal Consistency

A distinctive feature of the survey design is an internal consistency check (Question C.3), which compares the short-run empirical proportion Y_i/n_i with the vendor's recalled usage bucket. Vendors are then assigned a prior weight s_i according to the degree of agreement:

Consistency Level	Interpretation	s_i
Exact match	High reliability	10
Slight discrepancy	Moderate reliability	5
Major contradiction	Low reliability	2

This mechanism allows the model to adaptively down-weight recall responses whose implied usage rate conflicts sharply with the directly observed transaction behaviour.

11.1.4 Posterior Estimator

By conjugacy of the Beta–Binomial pair, the posterior is

$$p_i \mid Y_i \sim \text{Beta}(s_i m_i + Y_i, s_i(1 - m_i) + n_i - Y_i),$$

yielding the posterior mean estimator of vendor-level UPI intensity:

$$\hat{U}_i = \mathbb{E}[p_i \mid Y_i] = \frac{s_i m_i + Y_i}{s_i + n_i} = \underbrace{\frac{s_i}{s_i + n_i}}_{\text{recall weight}} m_i + \underbrace{\frac{n_i}{s_i + n_i}}_{\text{data weight}} \frac{Y_i}{n_i}.$$

The estimator is thus an adaptive weighted average of the recall-based estimate and the short-run empirical proportion, with weights determined jointly by the prior reliability and the volume of observed transactions. This formulation enjoys several desirable properties in the survey setting: it regularises estimates when n_i is small, avoids the degenerate boundary values 0 and 1 that arise from sparse observations, and mitigates recall bias by discounting internally inconsistent responses.

11.1.5 Advantages of the Bayesian Estimator

This approach is designed to handle the messiness of real-world data, where simple averages often fail us. One of the biggest challenges in tracking behavior is “small-sample noise.” For instance, if we only observe a few customers in an hour, a single person’s actions can wildly swing the percentages. This framework acts as a stabilizer; it keeps the estimates grounded so that a quiet hour doesn’t result in an exaggerated or misleading jump in the data. By looking at the bigger picture, the model can tell the difference between a genuine shift in behavior and a random, short-term fluke.

The framework also helps bridge the gap between what people say and what they actually do. We know from experience that when people describe their own habits, they often misremember details or give the answer they think we want to hear. This model is smart enough to spot those inconsistencies. If a customer’s reported habits don’t match the behavior we are seeing on the ground, the system automatically dials down the influence of those unreliable responses. This ensures that the final metrics are driven by actual evidence rather than faulty memory or “socially acceptable” answers.

Finally, in a typical raw count, if zero people use a digital tool during a specific window, the data might report a 0% adoption rate—which we know is rarely true in reality. This Bayesian approach avoids those unrealistic extremes of exactly zero or one hundred percent. Instead, it merges long-term trends with immediate observations to create a more nuanced, realistic measure of digital adoption.

11.1.6 Illustrative Example

Consider a vendor who selects the “6–8” recall category ($m_i = 0.70$), whose short-run proportion shows a slight discrepancy ($s_i = 5$), and who records $n_i = 4$ customers with $Y_i = 2$ UPI transactions in the preceding hour. The posterior mean is

$$\hat{U}_i = \frac{5(0.70) + 2}{5 + 4} = \frac{5.5}{9} \approx 0.611.$$

The estimate moderates the optimistic recall (0.70) toward the lower empirical rate (0.50), with the degree of shrinkage governed by the assigned reliability weight.

Chapter 12

Revenue Volatility Index

12.1. Bayesian Estimation of Revenue Instability

To quantify revenue volatility within the informal sector, this chapter employs a **Bayesian shrinkage** approach. Given the reliance on recall-based revenue extrema rather than longitudinal panels, this method regularizes observed weekly fluctuations by anchoring them to a vendor’s long-term business perception.

12.1.1 Observed Logarithmic Revenue Range

For each vendor i , we define the observed instability R_i using the highest ($X_{i,\max}$) and lowest ($X_{i,\min}$) daily revenues reported for the preceding week. We utilize a logarithmic ratio to ensure the measure is scale-free and representative of proportional fluctuations:

$$R_i = \log \left(\frac{X_{i,\max}}{X_{i,\min}} \right), \quad R_i \geq 0 \quad (12.1)$$

12.1.2 Likelihood and Prior Specifications

We model the observed range R_i as following a **Gamma distribution** characterized by a latent precision parameter ϕ_i . A higher ϕ_i indicates a more stable revenue stream.

- **Likelihood:** $R_i \mid \phi_i \sim \text{Gamma}(k, \phi_i)$, where k is a fixed shape parameter and ϕ_i is the rate.
- **Prior:** We leverage the vendor’s self-assessed stability category $s_i \in \{0, 1, 2\}$ to place a category-specific Gamma prior on the precision:

$$\phi_i \sim \text{Gamma}(\alpha_{s_i}, \beta_{s_i}), \quad \phi_i > 0 \quad (12.2)$$

12.1.3 Empirical Initialisation of Hyperparameters

The hyperparameters are initialized to align the prior with the empirical distribution of each stability category. Let $m_s = \text{median}(R_i \mid s_i = s)$ be the within-category sample median of the observed log-ranges. Based on the prior mean relationship $\mathbb{E}[R_i \mid \phi_i] = k/\phi_i$, we define a heuristic prior mean precision μ_s :

$$\mu_s = \frac{k}{m_s} \quad (12.3)$$

By introducing a scalar concentration parameter $\lambda > 0$ to govern prior strength, the shape and rate parameters are set as:

$$\alpha_s = \lambda\mu_s, \quad \beta_s = \lambda \quad (12.4)$$

This ensures that $\mathbb{E}[\phi_i \mid s_i = s] = \mu_s$, effectively encoding higher expected precision for vendors who self-report as more “steady.”

12.1.4 Posterior Distribution and the Instability Index

By conjugacy of the Gamma-Gamma model, the posterior distribution of the precision follows:

$$\phi_i \mid R_i \sim \text{Gamma}(\alpha_{s_i} + k, \beta_{s_i} + R_i) \quad (12.5)$$

We define the **Bayesian Instability Index** $\hat{\theta}_i$ as the reciprocal of the posterior mean precision:

$$\hat{\theta}_i = \frac{\beta_{s_i} + R_i}{\alpha_{s_i} + k} = \frac{\lambda + R_i}{\lambda\mu_{s_i} + k} \quad (12.6)$$

This estimator functions as a weighted average between the prior-implied instability and the weekly data, pulling volatile outliers toward the category mean while allowing the data to dominate as sample evidence grows.

12.2. Theoretical and Empirical Justification of the Bayesian Instability Index

The Bayesian Instability Index $\hat{\theta}_i$ developed in Section 12.1 departs deliberately from classical volatility estimators. In this section, we aim to benchmark the index against the Parkinson volatility estimator—the most natural classical comparator—and show, through empirical analysis, that the Bayesian index captures a richer informational structure that aligns more closely with vendors’ experienced instability.

12.2.1 Comparison with the Parkinson Volatility Estimator

The Parkinson Estimator and Its Adaptation to Survey Data

The Parkinson volatility estimator was developed in the financial econometrics literature as an efficient range-based measure of asset price volatility. Specifically, the Parkinson estimator is of the form:

$$\sigma_{\text{Parkinson}}^2 = \frac{1}{4 \ln 2} \left[\log \left(\frac{H}{L} \right) \right]^2, \quad (12.7)$$

where H and L denote the period's highest and lowest observed prices, respectively.

In the present study, the Parkinson estimator is adapted to the informal-sector revenue setting by substituting $H = X_{i,\max}$ and $L = X_{i,\min}$ —the highest and lowest daily revenues reported by vendor i over the preceding week—yielding:

$$P_i = \frac{1}{4 \ln 2} \left[\log \left(\frac{X_{i,\max}}{X_{i,\min}} \right) \right]^2. \quad (12.8)$$

Two important limitations attend this adaptation. First, the Parkinson estimator was derived for continuous-time, high-frequency stochastic processes; the present application uses a single week of recall-based data. There is no guarantee that the underlying revenue process satisfies the Brownian motion assumption, and the reported extrema are subject to recall bias. Second, and most critically, P_i is computed from a single observation period per vendor. There is therefore no opportunity to pool across time to reduce estimation error. The estimator should thus be regarded as a *descriptive analogue* of the Parkinson measure—capturing the same high-low range dispersion principle, but without the inferential guarantees of its financial econometric derivation.

These limitations motivate the Bayesian regularisation described in Section 12.1, and are directly addressed by the empirical comparisons that follow.

Empirical Comparison: Overall Correlation

Figure 12.1 presents the pooled scatterplot of $\hat{\theta}_i$ against P_i for all vendors in the sample. The relationship is positive and statistically significant, with a sample correlation of approximately $r \approx 0.60$. This moderate correlation is itself informative.

A correlation of $r \approx 0.60$ implies substantial shared variance ($R^2 \approx 0.36$), confirming that the Bayesian index is empirically grounded in the same underlying revenue fluctuation as the Parkinson measure. At the same time, roughly 64% of the variance in $\hat{\theta}_i$ is *not* explained by P_i . This residual variation is not noise: it reflects the structured contribution of the perceived-stability prior, which shifts each vendor's instability score toward the mean of their self-reported stability category.

Were the two measures perfectly collinear, the Bayesian construction would add no

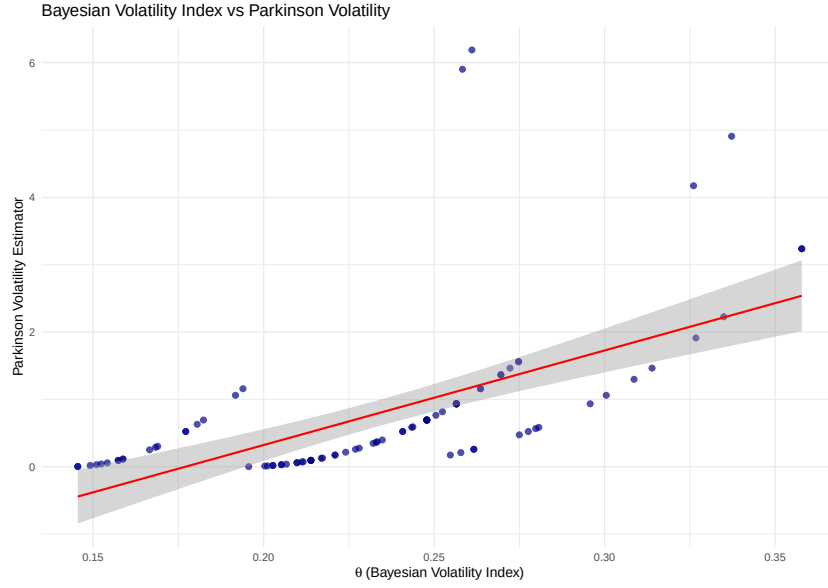


Figure 12.1. Pooled scatterplot of the Bayesian Instability Index $\hat{\theta}_i$ against the adapted Parkinson volatility estimator P_i . The fitted OLS regression line (shaded band: 95% confidence interval) indicates a positive but imperfect relationship ($r \approx 0.60$), suggesting that the two indices capture related but distinct aspects of revenue instability.

information beyond a monotone rescaling of the Parkinson range. The observed divergence, therefore, is precisely the empirical signature of the index's additional content.

Empirical Comparison: Behaviour Across Perceived-Stability Groups

The more revealing evidence comes from decomposing the relationship by perceived stability group. Figure 12.2 presents faceted scatterplots of $\hat{\theta}_i$ against P_i within each of the three self-reported stability categories: *Very Steady*, *Somewhat Fluctuating*, and *Very Unsteady*.

Several features of Figure 12.2 merit discussion.

Systematic group-level displacement. For any given value of P_i , the estimated $\hat{\theta}_i$ is systematically lower for *Very Steady* vendors and higher for *Very Unsteady* vendors. This vertical displacement across panels is a direct consequence of the category-specific denominator in equation (12.6): the same raw revenue range is mapped to a higher instability score when the vendor operates in a structurally volatile market. Economically, this captures the intuition that a daily revenue swing of, say, ₹500 is relatively unremarkable for a vendor who routinely experiences large fluctuations, but constitutes meaningful instability for a vendor with an otherwise stable clientele.

Within-group residual variation. Within each perceived-stability panel, the correlation between $\hat{\theta}_i$ and P_i remains positive but moderate, and the scatter is

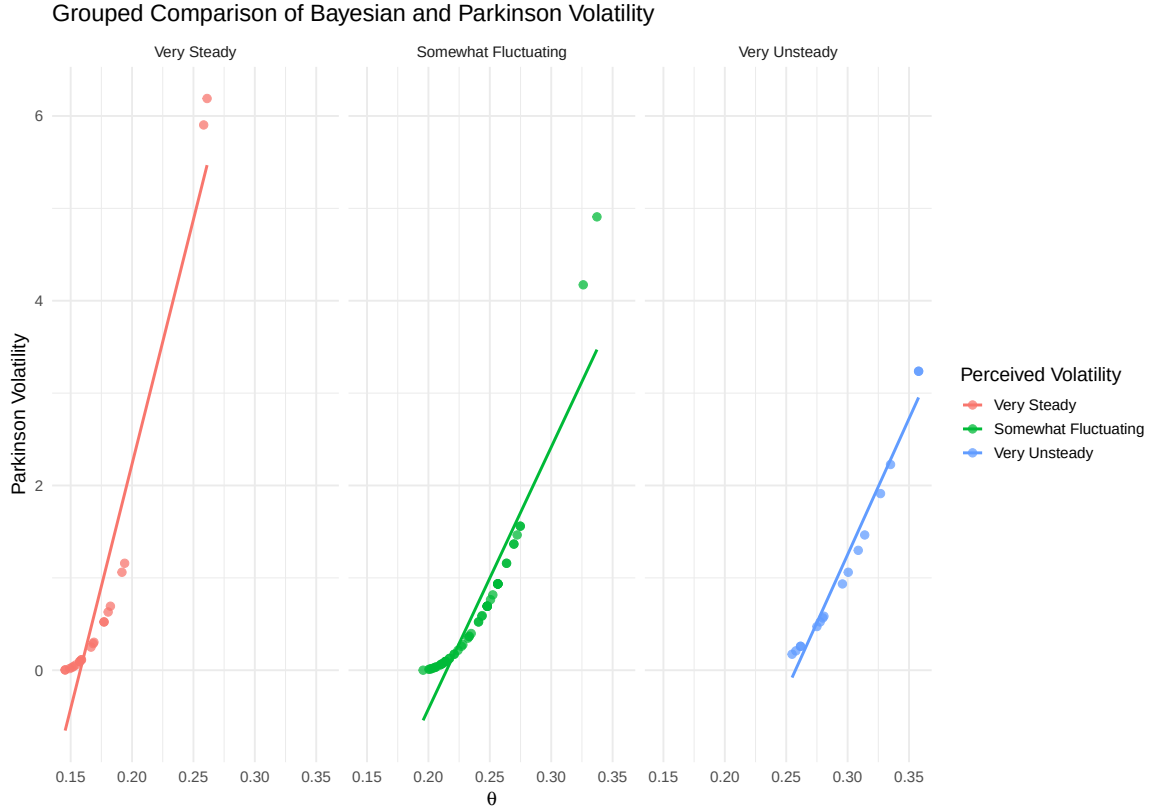


Figure 12.2. Faceted scatterplots of $\hat{\theta}_i$ against P_i , stratified by self-reported stability category. Within-group OLS regression lines are shown. The systematic vertical separation of the three panels indicates that the Bayesian index discriminates across perceived-stability groups conditional on a given level of raw range volatility.

substantial. This within-group variation arises because vendors in the same self-reported category differ in their category-median m_s , their individual R_i , and the extent to which the prior shrinks their posterior toward the group mean. The Bayesian index thus retains meaningful individual heterogeneity even after accounting for the group structure.

Critically, as shown in Figure 12.4, the distribution of P_i is nearly identical across the three perceived-stability categories—the interquartile ranges overlap substantially, and the group medians are close. By contrast, Figure 12.3 demonstrates that $\hat{\theta}_i$ separates the three groups cleanly, with a staircase-like progression from *Very Steady* to *Very Unsteady* vendors. The correlation between $\hat{\theta}_i$ and the ordinal perceived-stability variable is approximately $r \approx 0.79$, compared to near-zero for P_i .

Economic Interpretation

The pattern documented in Figures 12.2, 12.3 and 12.4 is consistent with a well-established distinction in the economic sociology of informal markets: *mechanical* revenue dispersion and *experienced* economic instability are related but distinct phenomena.

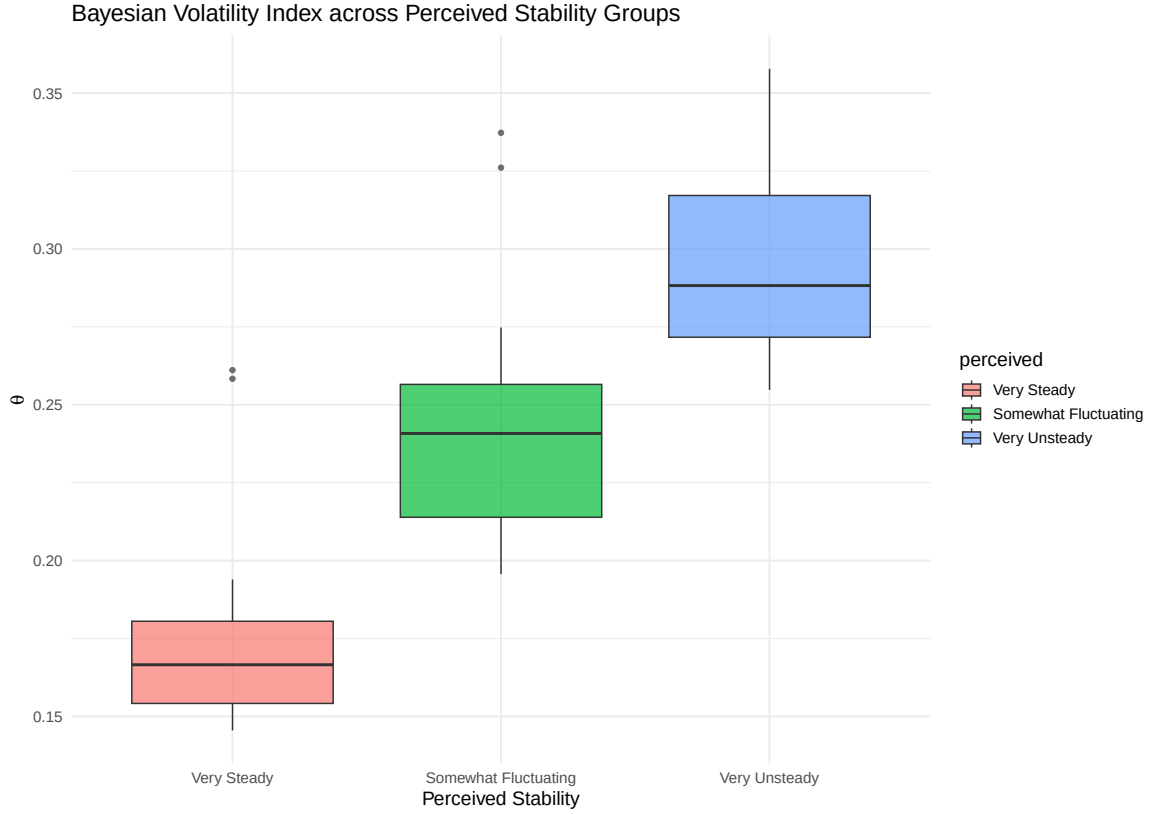


Figure 12.3. Bayesian Instability Index $\hat{\theta}_i$.

The Parkinson estimator measures mechanical dispersion—the raw range of revenue fluctuation over the observation period. Vendors may experience such fluctuations as routine if they arise from predictable demand cycles (weekly market days, seasonal footfall patterns), and such fluctuations do not necessarily constitute economically meaningful instability. Conversely, vendors may perceive high instability even when their within-week revenue range is moderate, if that range is accompanied by unpredictability in *which* days are high and which are low, uncertainty about future customer demand, or vulnerability to exogenous shocks.

The self-reported stability category s_i serves as a proxy for this broader experienced instability. By incorporating s_i into the prior, the Bayesian index regularises the raw range toward what is normative for vendors who share the same structural position in the market. The result is a measure that captures not merely how much revenue varied in a given week, but how that variation compares to the vendor’s typical operating environment—a distinction that is directly relevant to welfare analysis, credit access, and vulnerability mapping in informal economies .

Summary and Methodological Implications

The comparison between $\hat{\theta}_i$ and the adapted Parkinson estimator yields two conclusions that jointly justify the Bayesian construction.

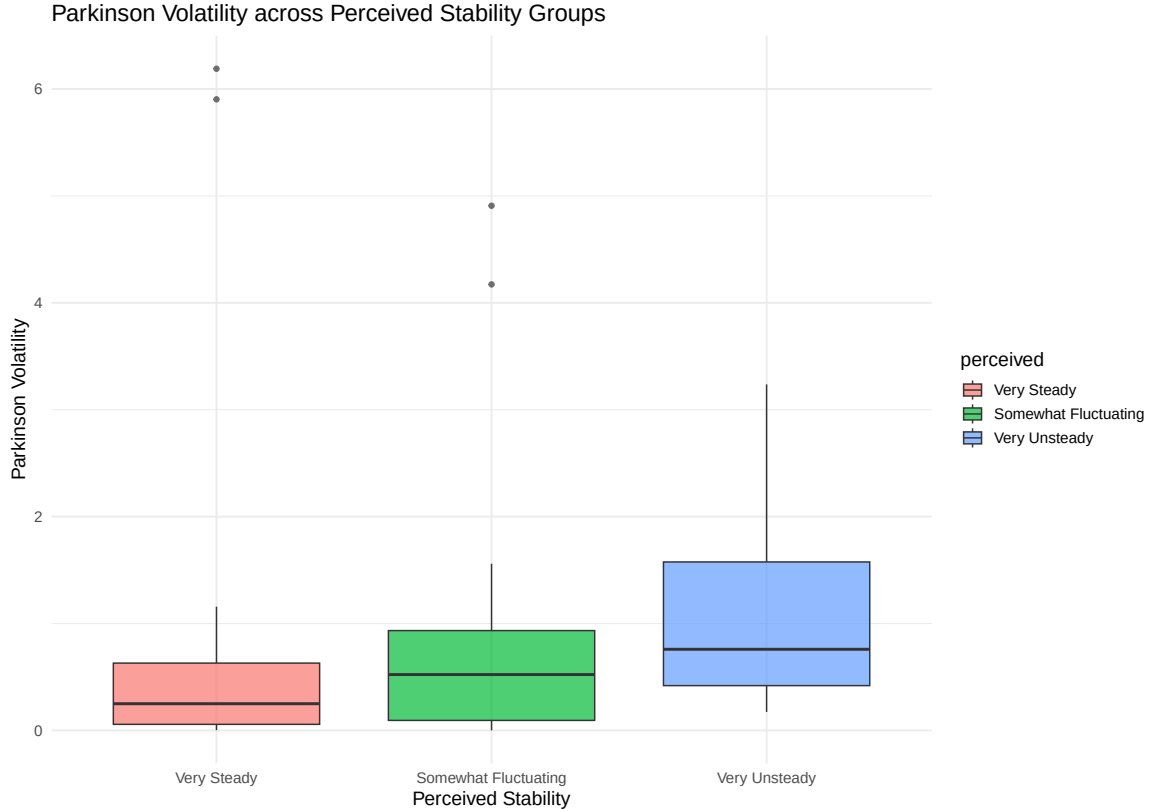


Figure 12.4. Parkinson volatility estimator P_i .

First, the two measures are positively correlated ($r \approx 0.60$), confirming that the Bayesian index is empirically grounded in observed revenue fluctuations and is not an artefact of prior specification.

Third, the group-level separation documented in Figures 12.2, 12.3 and 12.4 arises directly from the Bayesian shrinkage mechanism, which incorporates information about the vendor’s structural market position—information that is entirely discarded by the classical range estimator.

We emphasise that this does not constitute a claim that $\hat{\theta}_i$ is a superior volatility estimator in the statistical sense. The Parkinson estimator, applied to its intended high-frequency financial data, is theoretically optimal. Rather, the claim is that for the specific inferential task at hand—ranking informal vendors by their experienced economic instability using single-period survey data—the perception-regularised Bayesian index is a more informative and conceptually appropriate summary statistic. It functions as a *perception-regularised range volatility index*: objective enough to be anchored in observed revenue data, yet sensitive enough to the vendor’s structural context to separate economically meaningful instability from mechanical week-to-week fluctuation.

Chapter 13

UPI Adoption

13.1. Motivation and Objective

The primary objective of this inferential analysis is to determine whether vendors with different levels of UPI intensity differ significantly in their business characteristics and operational patterns. Since short-run observational counts are often highly variable and long-run recall responses may be affected by reporting bias, UPI intensity is measured using a reliability-weighted Bayesian index that combines both sources of information into a single statistically stable estimator.

In particular, the central question of interest is:

Is there a statistically significant relationship between UPI intensity and the socio-demographic or operational characteristics of street vendors?

The motivation behind this question is to understand whether digital-payment intensity among street vendors is primarily driven by business scale and operational factors, or whether the apparent patterns observed in raw transaction data are largely consequences of short-run variability and measurement noise.

13.2. Bayesian formulation of UPI intensity

Let n_i denote the total number of customers observed for vendor i during the observational window, and let Y_i denote the number of customers who used UPI for payment.

Conditional on the latent transaction intensity p_i , we assume

$$Y_i \mid p_i \sim \text{Binomial}(n_i, p_i).$$

The survey additionally records a coarse long-run estimate of UPI usage frequency. Let m_i denote the midpoint corresponding to the reported recall category. To combine the short-run observation with the long-run recall information, we impose the prior

$$p_i \sim \text{Beta}(s_i m_i, s_i(1 - m_i)),$$

where s_i represents a reliability weight determined through an internal consistency check between the observed transaction proportion and the recall response.

By Beta-Binomial conjugacy, the posterior distribution is again Beta, and the posterior mean estimator becomes

$$\hat{U}_i = \mathbb{E}(p_i \mid Y_i) = \frac{s_i m_i + Y_i}{s_i + n_i}.$$

Equivalently,

$$\hat{U}_i = \frac{s_i}{s_i + n_i} m_i + \frac{n_i}{s_i + n_i} \frac{Y_i}{n_i}.$$

Thus, the Bayesian estimator may be interpreted as a reliability-weighted average of:

1. the long-run recall-based estimate, and
2. the short-run empirical transaction proportion.

For the inferential analyses below, the Bayesian intensity index was binarized as follows:

$$\hat{U}_i \geq 0.5 \quad \Rightarrow \quad \text{High Intensity,}$$

$$\hat{U}_i < 0.5 \quad \Rightarrow \quad \text{Low Intensity.}$$

13.3. Questionnaire variables considered

The inferential analysis focuses on the following variables:

1. Age group of the vendor
2. Primary mode of operation
3. Business trajectory
4. Type of goods sold
5. Business duration

13.4. Analysis

Age group and UPI intensity

We first examined whether UPI intensity differs systematically across age groups. Earlier analyses based on raw intensity measures suggested a possible generational effect, so it is natural to ask whether this pattern persists after replacing the original index with the reliability-weighted Bayesian estimator.

Since both variables are categorical and the contingency table contains moderate cell frequencies, Pearson's Chi-squared test of independence was used.

The hypotheses considered were the following:

H_0 : Age group and UPI intensity are independent

H_1 : Age group and UPI intensity are associated

The contingency table and test results are shown below.

U_i Category	18–29	30–44	45–59	60+	Total
High (≥ 0.5)	8	27	22	5	62
Low (< 0.5)	5	24	24	14	67
Total	13	51	46	19	129

Table 13.1. UPI intensity vs. Age Group

Pearson's Chi-squared Test

Test statistic: $\chi^2 = 5.0327$

p-value: 0.1694

We fail to reject H_0 at the 5% significance level.

Although older vendors show somewhat higher counts in the low-intensity category, the overall distribution is not sufficiently different across age groups to establish statistical significance. Thus, once the Bayesian intensity measure is used, the apparent generational effect becomes substantially weaker.

Primary mode of operation and UPI intensity

We next investigated whether the physical mode of operation of the business influences transaction intensity. One might expect permanent stalls to display higher digital-payment adoption because of greater visibility and infrastructural stability. To test this, we compared UPI intensity across stalls, carts, and pavement-based setups.

Since some categories contain small counts, Fisher's Exact Test (Freeman–Halton extension for a 2×3 table) was used.

The hypotheses considered were the following:

H_0 : Mode of operation and UPI intensity are independent

H_1 : Mode of operation and UPI intensity are associated

The contingency table and test results are shown below.

U_i Category	Stall	Cart	Pavement	Total
High (≥ 0.5)	59	2	1	62
Low (< 0.5)	57	7	3	67
Total	116	9	4	129

Table 13.2. UPI intensity vs. Mode of Operation

Fisher's Exact Test

p-value: 0.2366

We fail to reject H_0 at the 5% significance level.

The analysis therefore does not provide evidence that the physical business setup materially influences transaction intensity. High-intensity usage appears achievable across all operational formats.

Business trajectory and UPI intensity

We also examined whether vendors reporting business growth differ in their UPI intensity from vendors reporting decline or stability. A plausible hypothesis is that growing businesses may adopt digital-payment systems more aggressively to handle larger or more dynamic transaction flows.

Since the expected cell frequencies were sufficiently large, Pearson's Chi-squared test of independence was employed.

The hypotheses considered were the following:

H_0 : Business trajectory and UPI intensity are independent

H_1 : Business trajectory and UPI intensity are associated

The contingency table and test results are shown below.

UPI Intensity	Grown	Declined	Same	Total
Low/Moderate	5	15	21	41
High	18	22	34	74
Total	23	37	55	115

Table 13.3. Business Trajectory vs. UPI Intensity

Pearson's Chi-squared Test

p-value: 0.2895

We fail to reject H_0 at the 5% significance level.

The data therefore do not indicate any statistically significant association between perceived business trajectory and transaction intensity.

Type of goods sold and UPI intensity

We next investigated whether the nature of goods sold influences digital-payment intensity. One possible intuition is that vendors selling perishable goods may rely more heavily on fast digital transactions because of higher turnover and continuous customer interaction.

The variables were analyzed using Pearson's Chi-squared test of independence.

The hypotheses considered were the following:

H_0 : Type of goods sold and UPI intensity are independent

H_1 : Type of goods sold and UPI intensity are associated

The contingency table and test results are shown below.

U_i Category	Perishable	Non-Perishable	Total
High (≥ 0.5)	27	35	62
Low (< 0.5)	39	28	67
Total	66	63	129

Table 13.4. UPI intensity vs. Type of Goods Sold

Pearson's Chi-squared Test

Test statistic: $\chi^2 = 2.2143$

p-value: 0.1367

We fail to reject H_0 at the 5% significance level.

Thus, the analysis does not provide sufficient evidence that the nature of goods sold plays a significant role in determining transaction intensity.

Business duration and UPI intensity

Finally, we examined whether the age of the business is associated with UPI intensity. Earlier exploratory analyses suggested that newer businesses may adopt digital payments more aggressively, whereas long-established businesses may retain stronger cash-based habits.

Since business duration is quantitative and no normality assumption was imposed, Mood's Median Test was used.

The hypotheses considered were the following:

H_0 : The median business duration is independent of UPI intensity

H_1 : The median business duration depends on UPI intensity

The grand median business duration in the analyzed sample was found to be 12 years.

The contingency table and test results are shown below.

U_i Category	≤ 12 Years	> 12 Years	Total
High (≥ 0.5)	35	23	58
Low (< 0.5)	28	36	64
Total	63	59	122

Table 13.5. Business Duration vs. UPI intensity

Mood's Median Test

p-value: 0.0989We fail to reject H_0 at the 5% significance level.

Although newer businesses appear somewhat more concentrated in the high-intensity category, the difference is not statistically significant at the conventional 5% level.

Operational characteristics and UPI intensity

While the earlier sections focused primarily on socio-demographic structure, a separate question is whether operational characteristics of the business are more strongly associated with UPI intensity. In particular, we examined whether vendors with higher transaction intensity differ systematically in customer volume and transaction value.

A natural expectation is that larger businesses with higher customer footfall would adopt UPI more aggressively. However, it is equally plausible that transaction value, rather than customer count, is the dominant driver of digital-payment intensity.

Customer volume and UPI intensity To compare customer volume across intensity groups, the Mann–Whitney U test was employed. Since daily customer counts are highly skewed and contain several extreme observations, a non-parametric rank-based procedure is more appropriate than a classical two-sample t -test.

The hypotheses considered were the following:

H_0 : The distribution of customer volume is the same across U_i groups

H_1 : The distributions differ significantly

The test results are summarized below.

Metric	U Statistic	p-value
Customer Avg Day	2335.0	0.0150

Table 13.6. Mann–Whitney Test for Customer VolumeMann–Whitney U Test**Inference:** Reject H_0 at the 5% significance level.

Interestingly, the direction of the effect is opposite to the naive expectation. The lower-intensity group ($U_i < 0.5$) actually occupies higher ranks for customer volume. In other words, vendors serving extremely large customer volumes are not necessarily the ones using UPI most intensively.

This pattern is reinforced by the logistic regression analysis discussed later, where customer volume appears with a negative coefficient. One plausible interpretation is

that vendors facing very high footfall may continue to prefer cash because it allows faster transaction handling during peak business hours.

Ticket size and UPI intensity We next investigated whether transaction value is associated with UPI intensity. Since ticket size distributions are similarly non-normal and skewed, the Mann–Whitney U test was again used.

The hypotheses considered were the following:

H_0 : The distribution of transaction value is the same across U_i groups

H_1 : The distributions differ significantly

The results are summarized below.

Metric	U Statistic	p-value
Ticket Size	2701.0	0.0023

Mann–Whitney U Test

Inference: Reject H_0 at the 5% significance level.

Table 13.7. Mann–Whitney Test for Ticket Size

Unlike customer volume, transaction value exhibits a strong positive relationship with UPI intensity. Vendors handling larger-value transactions are substantially more likely to belong to the high-intensity group.

Operationally, this is quite intuitive. Large cash transactions increase both logistical difficulty and security risk, making digital payments significantly more attractive. The evidence therefore suggests that transaction value acts as a stronger “pull factor” for UPI adoption than customer footfall itself.

Logistic regression analysis

To study the combined effect of multiple vendor characteristics simultaneously, a logistic regression model was fitted with UPI intensity ($U_i \geq 0.5$) as the response variable.

Unlike the earlier contingency-based analyses, the regression framework allows us to assess the partial effect of each variable while controlling for the others. Variance Inflation Factor (VIF) analysis was additionally performed to ensure that severe multicollinearity was absent among the predictors.

The overall model was statistically significant, with an LLR p-value of 0.0092.

The estimated coefficients are summarized below.

The regression results reinforce several earlier findings. Most socio-demographic variables remain statistically insignificant after adjustment. However, customer volume continues to display a significant negative association with UPI intensity, while business duration appears borderline significant.

Variable	Coefficient	p-value	VIF
Gender	0.6198	0.425	1.04
Age Group	-0.3556	0.182	1.14
Type of Goods Sold	0.4929	0.273	1.20
Business Duration	-0.0239	0.059	1.19
Mode of Operation	-0.7483	0.203	1.09
Customer Avg Day	-0.0142	0.039	1.30
Daily Amount Variability	-0.0016	0.996	1.10
Network Issue	-0.3270	0.210	1.15

Table 13.8. Logistic Regression Results

LLR p-value:

0.0092

All VIF values are comfortably below 5, indicating no serious multicollinearity.

Taken together, the evidence suggests that high digital-payment intensity is not simply a byproduct of larger customer volume. Instead, transaction profile and operational convenience appear to play a more important role in shaping sustained UPI adoption among street vendors.

As discussed with Prof. Debasis Sengupta, keeping variables such as Age in a binary form is appropriate, since splitting them into three sub-categories would result in insufficient data within each group. Therefore, the logistic regression model developed after making the necessary assumptions and combining categories to ensure fair representation across both ends does not require any modification.

13.5. Customer Arrival across UPI usage

We used *nonparametric tests* to compare customer arrival across UPI usage groups. This choice was appropriate because the outcome is discrete and potentially skewed, so normality-based methods are not ideal. Specifically, we used the **Wilcoxon rank-sum test** for comparing two groups, the **Kruskal–Wallis test** for comparing multiple groups, and the **Jonckheere–Terpstra test** for checking whether there is an ordered trend across UPI usage categories.

Second, to account for the count nature of the response and the presence of overdispersion, we fitted a **Negative Binomial regression model**. This model was preferred over a Poisson model because it allows the conditional variance to exceed the mean. It also permits adjustment for potential confounders such as age, gender, business duration, mode of operation, demand variability, and QR-code availability.

Summary of hypothesis tests

These results provide no strong evidence that customer arrival differs across the unadjusted UPI groups, nor do they indicate a clear monotonic trend.

Test	Statistic	df	p-value
Wilcoxon rank-sum test	$W = 1734$	NA	0.2573
Kruskal–Wallis test	$\chi^2 = 1.9327$	3	0.5865
Jonckheere–Terpstra test	$J = 2063.5$	3	0.2554

Table 13.9. Nonparametric tests for homogeneity across UPI groups.

Negative Binomial regression results

The regression analysis showed that most UPI-related variables were not statistically significant after adjustment. In particular, the coefficients for UPI acceptance and UPI frequency categories were not significant at conventional levels, suggesting that the raw association between UPI usage and customer counts is largely explained by other vendor characteristics.

Variable	Estimate	Std. Error	p-value
UPI Acceptance (Yes)	−0.377	0.839	0.653
UPI Frequency (3–5)	0.434	0.331	0.191
UPI Frequency (6–8)	−0.032	0.320	0.921
UPI Frequency (9–10)	−0.245	0.361	0.498

Table 13.10. Selected adjusted Negative Binomial regression coefficients for UPI-related variables.

Among the covariates, a few variables were significant, including age group, pavement mode, and unsteady demand variability, indicating that customer counts are better explained by vendor and operational characteristics than by UPI usage itself.

To check whether any UPI effect remained after controlling for confounders, nonparametric tests were applied to the adjusted residuals from the regression model.

Test	Statistic	df	p-value
Wilcoxon rank-sum test	$W = 1775$	–	0.042
Kruskal–Wallis test	$\chi^2 = 4.782$	3	0.188
Jonckheere–Terpstra test	$J = 1823$	–	0.101

Table 13.11. Nonparametric tests on adjusted residuals.

The Wilcoxon test shows a statistically significant difference between the low and high UPI groups, but this pattern does not persist in the Kruskal–Wallis or Jonckheere–Terpstra tests. Hence, the evidence for a genuine UPI effect is weak and not robust across different groupings.

Overall, the analysis suggests limited evidence of a systematic relationship between UPI usage and customer arrival. Although a weak difference appears when UPI is reduced to a binary comparison, the effect is not stable across finer groupings or trend-based tests. After adjusting for confounding variables in the Negative Binomial model, UPI-related effects become statistically insignificant, indicating that observed differences in

customer counts are driven mainly by vendor and operational characteristics rather than by UPI adoption itself.

13.6. Summary of findings

The use of a reliability-weighted Bayesian index substantially changes the inferential picture compared to analyses based on raw transaction intensity measures. After incorporating both recall-based and observational information into a statistically stabilized estimator, none of the examined socio-demographic or operational variables exhibit statistically significant association with UPI intensity at the 5% significance level.

This suggests that several apparent patterns observed under simpler intensity measures may largely reflect short-run variability or measurement noise rather than stable structural relationships. Overall, the analysis indicates that digital-payment adoption among street vendors is more heterogeneous and less strongly tied to observable demographic characteristics than initially expected.

Chapter 14

Cash Change Shortage and UPI Adoption

14.1. Motivation and Objective

Cash is still central to everyday street vending, but small cash transactions often create a very specific friction: the vendor may not have enough change to complete the sale smoothly. This problem is easy to overlook because each individual loss looks small. Across repeated transactions, however, it can become a meaningful source of lost revenue.

The objective of this chapter is therefore twofold:

To quantify the magnitude of sales lost because of the cash change-shortage problem, and to examine whether higher UPI usage mitigates this loss.

The chapter treats the change-shortage problem not merely as an inconvenience, but as an observable transaction failure. If a customer walks away because exact change is unavailable, the vendor loses the sale that would otherwise have taken place. UPI can reduce this friction because it allows the payment to be completed without depending on cash denominations.

14.2. Variable Construction and Data Preparation

The analysis uses the cleaned vendor survey data. One extreme transaction-size observation of Rs. 200000 was removed because it was far outside the remaining distribution

and would have dominated both the descriptive statistics and the regression estimates. After cleaning, the revenue-loss calculation is based on 118 usable observations.

Several responses were recorded in interval form. These were converted into numerical midpoints so that they could be used in quantitative analysis. For example, a reported interval of “3–5 customers” was treated as 4 customers. This midpoint approach is imperfect, but it provides a transparent way of turning categorical survey responses into approximate numerical measures.

14.3. Revenue Loss Monetisation

Let

CLC_i = estimated number of customers lost due to cash shortage

and

TS_i = average transaction size for vendor i .

The estimated daily revenue loss for vendor i is defined as

$$RL_i = CLC_i \times TS_i.$$

The customer-loss intervals were mapped into midpoint values as follows:

Table 14.1. Midpoint conversion for customer-loss intervals

Reported interval	Midpoint used
0–2	1.0
3–5	4.0
6–8	7.0
9–10	9.5

This construction gives an approximate monetary measure of the daily sales that vendors lose because of the inability to provide cash change.

14.4. Descriptive Results

The distribution of estimated revenue loss is highly uneven. Most vendors report relatively modest losses, but a smaller group reports very large losses. This is reflected in the gap between the mean and the median.

Table 14.2. Summary statistics for estimated daily revenue loss

Statistic	Value
Observations	118
Median	240.00
Mean	759.92
Standard deviation	2143.27
Minimum	0.00
Maximum	15000.00
25th percentile	60.00
75th percentile	300.00
Skewness	4.67
Excess kurtosis	22.92
Annualised median loss	87600.00

The median daily loss is Rs. 240. On an annualised basis, using a simple daily extrapolation, this corresponds to Rs. 87600. The mean loss is much higher because the distribution is strongly right-skewed. For this reason, the median is the more reliable measure of a typical vendor's experience.

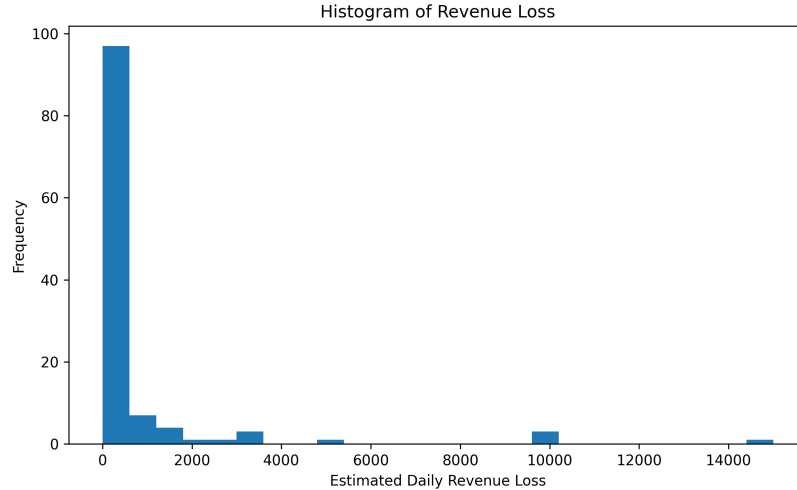
**Figure 14.1.** Histogram of estimated daily revenue loss

Figure 14.1 shows that the distribution is concentrated near the lower end, with a long right tail. The log-transformed histogram in Figure 14.2 makes the spread among smaller and medium-sized losses easier to see.

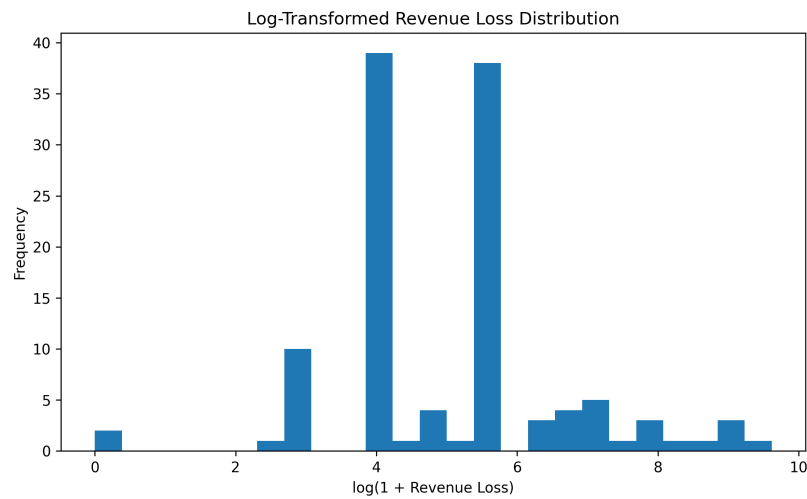


Figure 14.2. Log-scale histogram of estimated daily revenue loss

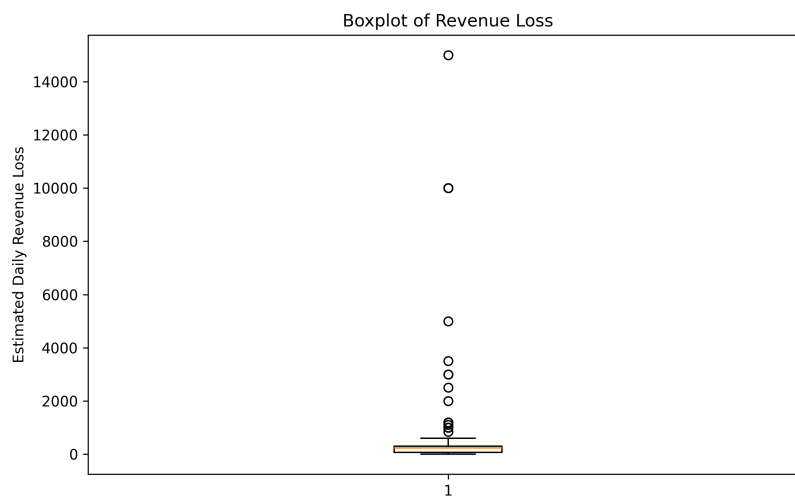


Figure 14.3. Boxplot of estimated daily revenue loss

The boxplot further confirms the presence of extreme observations. These are not necessarily errors: for higher-ticket vendors, even a small number of lost customers can translate into large monetary losses.

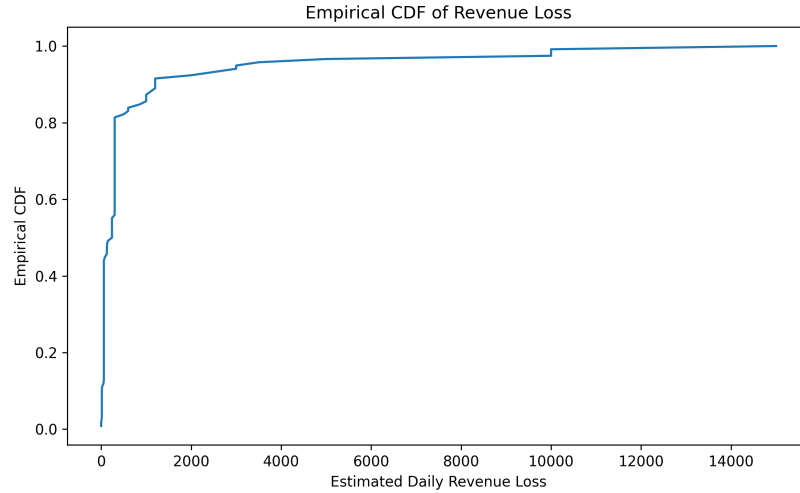


Figure 14.4. Empirical cumulative distribution function of estimated revenue loss

The empirical cumulative distribution function indicates that the majority of vendors face losses clustered below a few hundred rupees per day, while the top tail carries a disproportionate share of the total monetary burden.

14.5. UPI Intensity as a Mitigation Mechanism

A simple yes/no measure of UPI adoption is not very informative in this dataset because UPI adoption is already common. The more relevant question is whether vendors use UPI intensively enough for it to matter in daily operations.

To capture this, the analysis uses a Bayesian UPI intensity estimator. Recall-based UPI usage was first converted into prior mean values:

Table 14.3. Prior mean construction for UPI usage intervals

UPI usage interval	Prior mean
0–2	0.10
3–5	0.40
6–8	0.70
9–10	0.95

The final estimator combines the prior usage information with observed short-run customer transactions:

$$\hat{U}_i = \frac{s_i m_i + Y_i}{s_i + n_i},$$

where m_i is the prior UPI intensity estimate, s_i is a reliability weight, Y_i is the number of observed UPI-paying customers, and n_i is the observed number of total customers in the relevant short-run window.

This construction is useful because it does not treat all UPI adopters as identical. A vendor who accepts UPI once in a while and a vendor who completes most sales through UPI are placed at different points on the intensity scale.

14.6. Regression Framework

The original customer-loss categories were sparse in the upper tail, so they were collapsed into a binary outcome:

Table 14.4. Binary outcome definition for customer-loss severity

Reported customer-loss category	Binary outcome
0–2 customers lost	0: Low loss
3 or more customers lost	1: Elevated loss

The logistic regression model is

$$\log \left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)} \right) = \beta_0 + \beta_1 \hat{U}_i + X_i^\top \beta,$$

where $Y_i = 1$ denotes elevated customer-loss severity and X_i includes cash dependence, ticket size, business duration, goods type, and daily revenue variability.

Because the low-loss category is much larger than the elevated-loss category, both unweighted and inverse-frequency weighted logistic regressions were estimated. The weighted model is emphasized because it gives greater importance to the elevated-loss observations, which are the main focus of the mitigation question.

14.7. Regression Results

Table 14.5. Binary logistic regression results

Predictor	Coef.	SE	z	p -value	Odds ratio
Intercept	-0.7375	1.3403	-0.5502	0.5822	0.4783
Cash dependence	-0.1596	0.1621	-0.9845	0.3249	0.8525
Ticket size	-0.0035	0.0026	-1.3065	0.1914	0.9965
Business duration	0.0300	0.0161	1.8601	0.0629	1.0304
Goods type	0.1170	0.6788	0.1723	0.8632	1.1241
Bayesian UPI intensity $\hat{U}_i$	-2.4118	0.9449	-2.5525	0.0107	0.0897
Daily variability 1	0.4860	0.5328	0.9123	0.3616	1.6259
Daily variability 2	1.1986	0.9470	1.1257	0.2056	3.3154

In the unweighted model, the coefficient on $\hat{U}_i$ is negative and statistically significant. Its odds ratio is 0.0897, meaning that higher UPI intensity is associated with a much lower probability of elevated customer loss. Because the UPI intensity metric ($\hat{U}_i$) is bounded between 0 and 1, a full unit increase in the model would represent an unrealistic jump from a completely cash-based business to one that is entirely digital. To ground our findings in a more practical context, we evaluate the impact of a 10-percentage-point increase in digital adoption. By scaling the coefficient accordingly, we find that the odds ratio for a 0.1 increase in $\hat{U}_i$ (calculated as $e^{0.1 \cdot \hat{\beta}_1}$) is approximately 0.78574. To look at it from another perspective, the odds of experiencing customer loss due to cash change shortage are roughly 1.27 times higher for businesses with lower digital engagement. This suggests that a lack of digital payment options carries a 27% “friction penalty” in terms of potential customer attrition.

Table 14.6. Weighted binary logistic regression results

Predictor	Coef.	SE	z	p -value	Odds ratio
Intercept	0.6899	0.5869	1.1754	0.2398	1.9934
Cash dependence	-0.1238	0.1009	-1.2268	0.2199	0.8835
Ticket size	-0.0039	0.0031	-1.2889	0.1974	0.9961
Business duration	0.0399	0.0124	3.2046	0.0014	1.0407
Goods type	0.3162	0.5835	0.5419	0.5879	1.3719
Bayesian UPI intensity $\hat{U}_i$	-2.3367	0.7840	-2.9805	0.0029	0.0966
Daily variability 1	0.5992	0.6635	0.9031	0.3665	1.8207
Daily variability 2	1.3612	0.7186	1.8942	0.0582	3.9009

The weighted model gives the clearest evidence for the mitigation hypothesis. The

coefficient on UPI intensity remains negative and becomes highly significant. The odds ratio of approximately 0.10 implies that vendors with greater UPI intensity are substantially less likely to report elevated customer loss from cash change shortages.

Ticket size is also significant in the weighted model, with a negative coefficient. This suggests that larger-ticket vendors are somewhat less likely to suffer elevated customer loss after controlling for other factors. Business duration and daily variability are positively associated with elevated loss, implying that older businesses and more unstable daily revenue environments face greater exposure to transaction frictions.

Table 14.7. Model comparison

Metric	Unweighted model	Weighted model
Classification accuracy	0.8843	0.7025
McFadden pseudo R^2	0.1596	0.5128
AUC	0.7755	0.7836

The unweighted model has higher classification accuracy, but this is partly because it can perform well by predicting the majority low-loss category. The weighted model has a much higher McFadden pseudo R^2 and a slightly higher AUC, which makes it more useful for understanding elevated-loss cases.

14.8. Interpretation

The main result is that UPI adoption matters most when it is actually used intensively. Merely having the option to accept UPI is not enough. The mitigating effect appears when UPI becomes a regular part of the vendor's transaction process.

This supports a practical interpretation: UPI reduces the dependence on exact cash availability. When customers can pay digitally, the sale is less likely to fail because of change shortages. The result is especially relevant for vendors whose daily business involves many small payments, where even a minor shortage of coins or small notes can disrupt sales.

14.9. Overall Conclusion

The cash change-shortage problem imposes a measurable cost on vendors. The typical estimated loss is modest on a daily basis, but it becomes economically meaningful when repeated over time. The analysis also shows that higher UPI intensity is strongly

associated with lower customer-loss severity. Thus, UPI appears to play a real mitigating role, not simply by being available, but by being integrated into everyday selling.

Chapter 15

Daily Revenue Volatility and UPI Adoption

15.1. Motivation and Objective

A second important question is whether UPI stabilizes vendor income. If digital payments reduce small transaction frictions, broaden the set of customers who can pay, and reduce dependence on cash availability, then vendors who use UPI more intensively may be expected to face less day-to-day revenue instability.

The objective of this chapter is:

To estimate daily revenue volatility and determine whether UPI intensity is negatively correlated with revenue volatility, after controlling for goods type and location.

The analysis in this chapter uses a *Bayesian shrinkage instability index* as the revenue volatility measure. This is a regularised index built from the highest and lowest reported daily revenues in the preceding week, together with the vendor's self-reported stability category. In the discussion below, this is treated as the **revenue volatility index** for the chapter.

15.2. Revenue Volatility Index Used in the Analysis

The survey does not provide a full daily revenue panel. Instead, each vendor reported the highest and lowest daily earnings in the preceding seven days. For vendor i , let

H_i denote the highest reported daily earning and L_i denote the lowest reported daily earning. The observed logarithmic revenue range is

$$R_i = \log \left(\frac{H_i}{L_i} \right), \quad R_i \geq 0.$$

This range is scale-free: a gap from Rs. 500 to Rs. 1,000 and a gap from Rs. 5,000 to Rs. 10,000 receive the same log-range value because both represent a doubling of revenue. This is useful in a sample where vendors operate at very different revenue levels.

However, R_i is still noisy because it is based on a one-week recalled range rather than repeated daily observations. The Q3 analysis therefore regularises this measure using each vendor's self-reported long-run stability category. Let

$$s_i \in \{0, 1, 2\}$$

represent whether the vendor describes daily revenue as very steady, somewhat fluctuating, or very unsteady.

The model treats the observed log-range as Gamma-distributed conditional on a latent precision parameter:

$$R_i \mid \phi_i \sim \text{Gamma}(k, \phi_i),$$

where a larger value of ϕ_i corresponds to more stable revenue. A category-specific Gamma prior is then assigned to ϕ_i :

$$\phi_i \sim \text{Gamma}(\alpha_{s_i}, \beta_{s_i}).$$

Using conjugacy, the posterior distribution is

$$\phi_i \mid R_i \sim \text{Gamma}(\alpha_{s_i} + k, \beta_{s_i} + R_i).$$

The Bayesian shrinkage instability index is defined as the reciprocal of the posterior mean precision:

$$\hat{\theta}_i = \frac{\beta_{s_i} + R_i}{\alpha_{s_i} + k}.$$

This is the index used throughout the chapter. A larger value of $\hat{\theta}_i$ indicates greater revenue instability. The term “shrinkage” is important: the index does not simply reproduce the raw best-day/worst-day range. Instead, it pulls each vendor's observed weekly range toward the average instability associated with their stated stability category. This makes the measure less sensitive to unusually high or low recalled values.

In the baseline Q3 specification, $k = 5$ and the prior concentration parameter is set at

$\lambda = 5$. The prior hyperparameters are obtained from the empirical median log-ranges within the three stability categories. If m_s denotes the median value of R_i in category s , then the prior mean precision is initialised as

$$\mu_s = \frac{k}{m_s},$$

and the hyperparameters are set as

$$\alpha_s = \lambda\mu_s, \quad \beta_s = \lambda.$$

Thus, the index may also be written as

$$\hat{\theta}_i = \frac{\lambda + R_i}{\lambda\mu_{s_i} + k}.$$

This is best interpreted as a regularised revenue instability index rather than a literal estimate of statistical variance. It is nevertheless appropriate for this survey because it uses exactly the available information: recalled revenue extrema and vendors' own assessment of business stability.

15.3. Data and Descriptive Statistics

The full survey contains 129 vendors. UPI intensity is available for all vendors, while valid highest and lowest revenue responses are available for 92 vendors. These 92 observations form the main analytical sample for the volatility exercise.

Table 15.1. Descriptive statistics for revenue instability analysis

Variable	n	Mean	Median	SD	Min	Max
UPI intensity U_i	129	0.504	0.467	0.284	0.000	0.971
Log revenue range R_i	92	1.216	1.204	0.859	0.041	4.143
Bayesian instability index $\hat{\theta}_i$	92	0.234	0.238	0.049	0.146	0.358
Non-perishable goods indicator	129	0.488	—	—	0	1
UPI accepted	129	0.930	—	—	0	1

The first validity check is whether the index follows the ordering implied by vendors' own responses. It does. Among vendors with valid revenue-extrema responses, the mean value of $\hat{\theta}_i$ rises from 0.174 for very steady vendors to 0.238 for somewhat fluctuating vendors and 0.297 for very unsteady vendors.

Table 15.2. Bayesian instability index by self-reported stability category

Stability category	Valid n	Mean of $\hat{\theta}_i$	Median of $\hat{\theta}_i$
Very steady	21	0.174	0.167
Somewhat fluctuating	55	0.238	0.241
Very unsteady	16	0.297	0.288

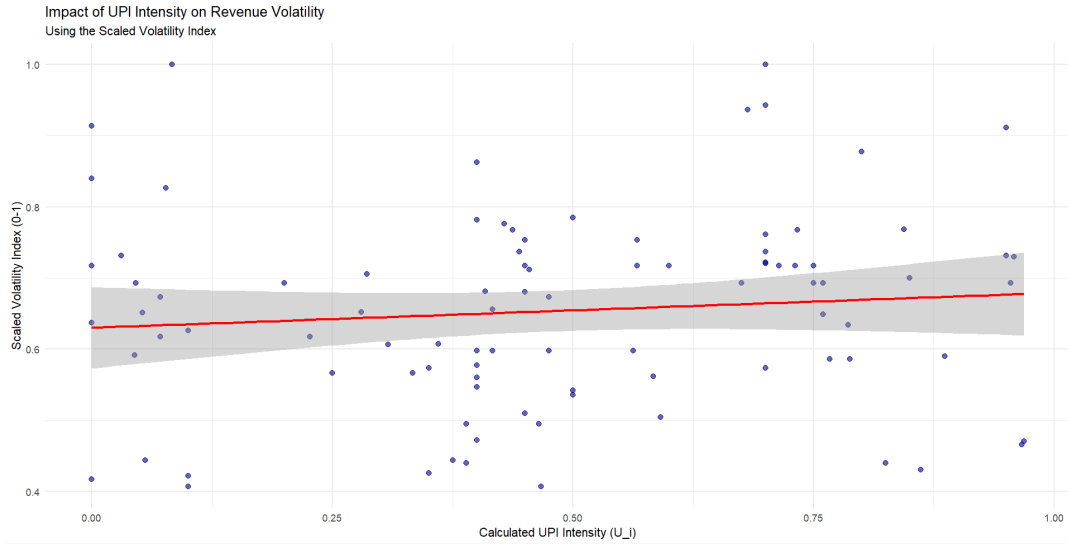
A Kruskal–Wallis test strongly rejects equality of the index distributions across these three stability categories:

$$H = 54.36, \quad p < 0.0001.$$

This supports the internal validity of the measure: vendors who describe their revenues as more unstable are also assigned higher values of the Bayesian instability index.

15.4. Exploratory Plots

The aggregate relationship between UPI intensity and scaled revenue volatility is weak and slightly positive in the exploratory plot. This already cautions against a simple interpretation in which more UPI use automatically produces more stable revenue.

**Figure 15.1.** Overall relationship between UPI intensity and scaled revenue volatility

A more informative view comes from segmenting vendors by ticket size. Higher-ticket stalls show higher volatility, which is consistent with lumpy demand. A vendor selling low-priced items may have many small transactions in a day, while a high-ticket vendor may have a few large sales or none at all.

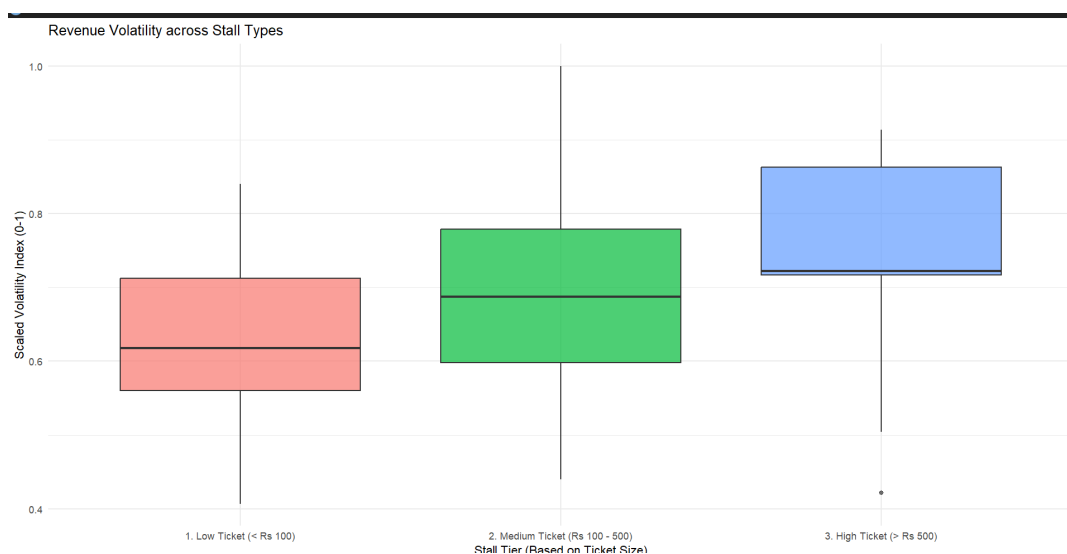


Figure 15.2. Revenue volatility across ticket-size groups

When UPI intensity is plotted separately by stall tier, the relationship appears heterogeneous. The low-ticket group shows a weak negative slope, while the medium- and high-ticket groups do not show a stabilising pattern.

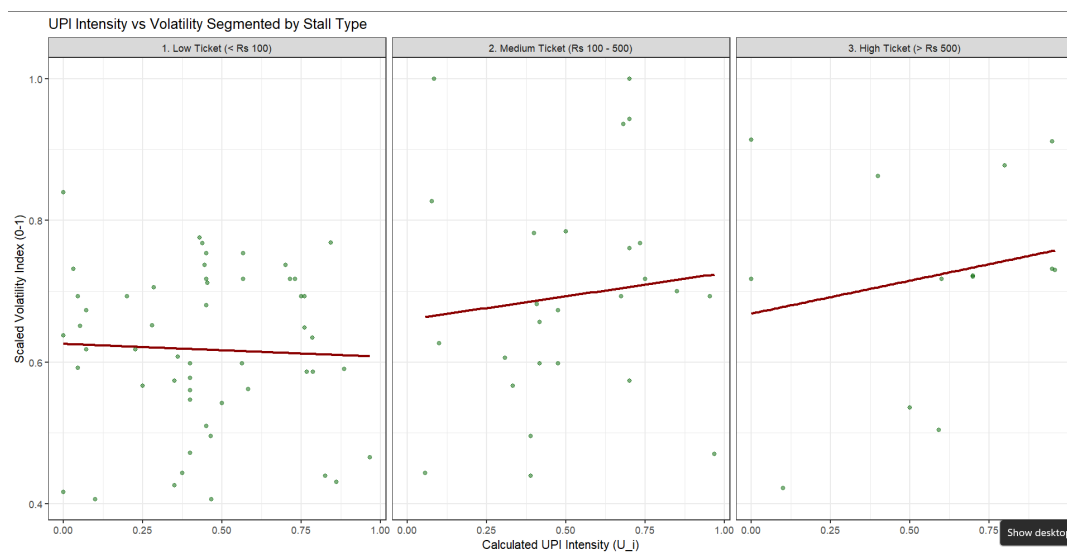


Figure 15.3. UPI intensity and volatility segmented by ticket-size group

This pattern is economically plausible. For low-ticket vendors, the exact-change problem is a frequent operational friction. UPI can help complete small transactions that may otherwise be delayed or lost. For high-ticket vendors, however, instability is more likely to come from irregular demand rather than from the payment mode itself.

15.5. Hypothesis Testing

15.5.1 Bivariate Relationship

The Spearman rank correlation between UPI intensity and the Bayesian instability index is

$$\rho_S = 0.135, \quad p = 0.200.$$

The sign is positive rather than negative, and the result is not statistically significant. A Mann–Whitney comparison between vendors below and above the median UPI intensity gives

$$U = 887.0, \quad p = 0.909.$$

The median instability is 0.224 for the low-UPI group and 0.248 for the high-UPI group. This again gives no evidence of a general stabilising effect of UPI intensity on revenue volatility.

15.5.2 Sensitivity Analysis

The Q3 analysis checks whether the UPI result depends on the Bayesian hyperparameters. Across $k \in \{3, 5, 10\}$ and $\lambda \in \{2, 5, 10\}$, the UPI coefficient remains small and statistically insignificant in all nine specifications. The p -values are always greater than 0.30. Hence, the null finding is not driven by one particular choice of hyperparameters.

15.6. Overall Conclusion

The results do not show evidence that UPI intensity reduces overall volatility. The estimated UPI effect is small, statistically insignificant, and not in the expected negative direction.

The stronger finding is that revenue instability is shaped by the structure of the business itself. Goods type and ticket size matter because they determine whether revenue arrives through many small transactions or through fewer, more irregular sales. UPI may still help low-ticket vendors by reducing exact-change frictions, as suggested by the segmented exploratory plot. But at the full-sample level, the evidence points more toward demand structure than payment intensity as the main driver of daily revenue volatility.

Chapter 16

Vendor Perception of UPI

16.1. Motivation and Objective

The preceding chapters examine UPI through business outcomes: customer volume, revenue loss, and revenue volatility. This chapter turns to vendor perceptions. Perceptions matter because a payment system is not adopted only on technical grounds. Vendors also evaluate whether it is faster, easier to trust, useful for keeping money in the bank, and reliable enough for daily selling.

The objective of this chapter is:

To characterise vendor perceptions of UPI with respect to transaction speed, financial tracking, savings behaviour, and digital trust, segmented by goods type, age group, and mode of operation.

The analysis relies mainly on contingency tables and exact tests because several response categories have small counts.

16.2. Variables and Statistical Approach

The perception-related variables used in this chapter include:

- perceived transaction speed of UPI relative to cash,
- whether UPI helps vendors save more because money goes directly to the bank,
- worry about digital records, security, or trust,
- network issues and mode preference,
- customer loss due to UPI-related problems.

These responses were analysed against age group, goods type, and primary mode of operation. Fisher’s Exact Test was used whenever expected cell counts were small; otherwise Pearson’s Chi-square test was used. The 5% level is used as the reference threshold throughout.

16.3. Transaction Speed

The first perception question asks whether UPI is faster, slower, or roughly the same as cash. The table below cross-tabulates this response with age group.

Table 16.1. Age group and perceived transaction speed of UPI

Time to UPI	18–29	30–44	45–59	60+	Total
Faster	7	32	31	10	80
No change	3	8	7	3	21
Slower	0	1	2	2	5
Total	10	41	40	15	106

Fisher’s Exact Test gives

$$p = 0.668.$$

Thus, perceived transaction speed is not significantly associated with age. The main descriptive point is stronger than the inferential one: across all age groups, most vendors say that UPI is faster. This suggests that speed is a widely recognised advantage of UPI, rather than a perception held only by younger vendors.

16.4. Savings Behaviour and Financial Tracking

The survey also asks whether vendors feel they save more because UPI payments go directly to the bank. This question captures a broader idea of financial tracking: if digital payments remain in the bank account, vendors may find it easier to monitor receipts or avoid immediate cash spending.

The response distribution was tested against age group, goods type, and mode of operation.

Table 16.2. Savings behaviour response by age group

Response	18–29	30–44	45–59	60+
Yes	4	15	9	3
No	2	12	14	4
No difference	2	13	16	6

The Fisher test for age group gives $p = 0.7167$. Similar tests for goods type and mode of operation also fail to show significant association.

Table 16.3. Tests for association with perceived UPI savings behaviour

Segmentation variable	Test used	p -value
Age group	Fisher’s Exact Test	0.7167
Goods type	Pearson Chi-square	0.8188
Primary mode of operation	Fisher’s Exact Test	0.7803

These results suggest that views on UPI and savings are broadly similar across demographic and operational groups. Some vendors do see UPI as helping savings or record-keeping, but this perception is not concentrated in a particular age group, goods category, or mode of operation.

16.5. Digital Trust, Network Issues, and Mode Preference

Digital trust was studied through concerns about digital records or security. The table below shows security concern by age group.

Table 16.4. Age group and digital security concern

Security concern	18–29	30–44	45–59	60+	Total
No worry	6	22	27	10	65
Yes, worried	1	6	4	2	13
Neutral	3	12	8	3	26
Total	10	40	39	15	104

Fisher’s Exact Test gives

$$p = 0.928.$$

There is no evidence that security concern varies by age. This is useful because it challenges a common assumption that older vendors are necessarily more distrustful of digital payments. In this sample, the proportion reporting no worry is high across all age groups.

The relationship between network issues and payment-mode preference was also tested.

Table 16.5. Network issues and preferred payment mode

Network issues	Prefers UPI	Prefers cash	Neutral	Total
Daily	1	4	2	7
Weekly	5	6	5	16
Monthly	5	6	3	14
Rarely	31	21	23	75
Total	42	37	33	112

The exact test gives $p = 0.669$. Even vendors who report more frequent network issues are not statistically more likely to prefer cash. This does not mean network issues are irrelevant, but it suggests that occasional technical friction has not erased the perceived usefulness of UPI.

16.6. Reliability and Business Experience

A related question is whether UPI-related customer loss is connected to mode of operation or business trajectory. Two contingency analyses were conducted.

Table 16.6. Mode of operation and customers lost to UPI issues

Mode of operation	Never	Sometimes	Often	Total
Permanent stall	77	15	4	96
Mobile cart	4	3	0	7
Pavement	2	0	0	2
Total	83	18	4	105

The Fisher test gives $p = 0.3398$, so customer loss due to UPI issues is not significantly associated with the physical mode of operation.

Table 16.7. Business trajectory and customers lost to UPI issues

Business trajectory	Never	Sometimes	Often	Total
Grown	17	5	0	22
Declined	34	4	0	38
Same	32	9	4	45
Total	83	18	4	105

The Fisher test gives $p = 0.1219$. This is not significant at the 5% level, but the descriptive pattern is worth noting: all vendors who often lost customers due to UPI issues reported that their business trajectory was the same, not grown. This is not conclusive evidence of stagnation, but it suggests that repeated digital friction may limit upside even if it does not cause immediate decline.

A Kruskal–Wallis test also found that high earnings over the last seven days differ significantly by customer-loss category:

$$p = 0.0266.$$

The corresponding test for low earnings was not significant ($p = 0.3375$). This suggests that UPI-related friction may matter more for best-day performance than for minimum daily earnings.

16.7. Overall Conclusion

The perception results show a broad acceptance of UPI’s practical value. Most vendors view UPI as faster than cash, and this perception is not limited to younger vendors. Savings-related perceptions are more mixed, but they do not vary significantly by age, goods type, or mode of operation. Trust concerns also do not show a meaningful age gradient.

The most important takeaway is that vendors’ perceptions of UPI are relatively uniform across groups. UPI appears to have become a normal part of informal market transactions. Differences exist in individual experience, especially where network or transaction failures occur, but the overall pattern is not one of strong demographic or operational divides in perception.

Chapter 17

Motivations behind UPI Adoption

17.1. Motivation and Objective

The final research question concerns why vendors adopt UPI and what fee structures they operate under. Since UPI adoption is already widespread in the sample, the more useful question is not simply who adopts, but what pushes vendors toward adoption and what arrangements shape their use of digital payment infrastructure.

The objective of this chapter is:

To identify the primary motivating factors behind UPI adoption and the structural fee arrangements vendors operate under.

This chapter has two linked components. First, it studies the reasons vendors report for adopting UPI. Second, it studies the fee arrangements associated with UPI use, particularly whether vendors operate with no rent, device rent, or per-sale rent.

17.2. Primary Motivating Factors

The most common motivation for UPI adoption is customer demand. This is a crucial point: many vendors did not adopt UPI because of abstract enthusiasm for digital finance, but because customers increasingly expected it.

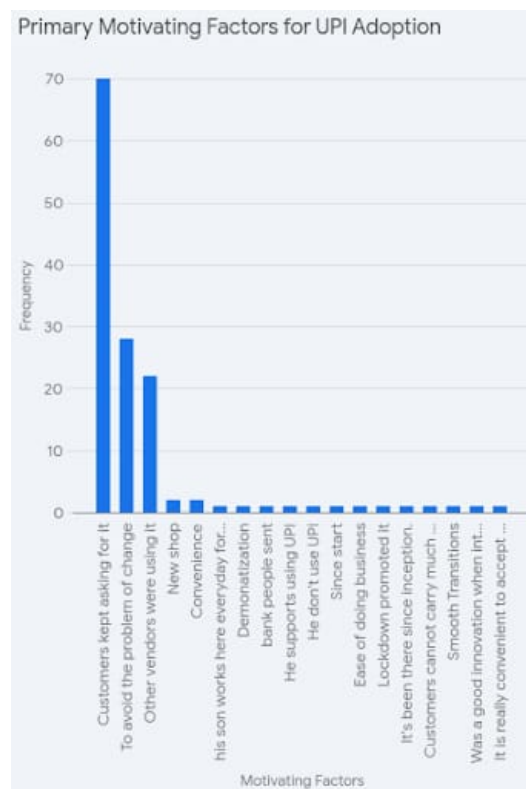


Figure 17.1. Primary motivating factors for UPI adoption

The main reported motivations are:

Table 17.1. Primary motivating factors for UPI adoption

Motivating factor	Number of vendors
Customers kept asking for it	70
To avoid the problem of change	28
Other vendors were using it	22
Other secondary reasons	Small counts

The pattern is clear. Customer demand is the dominant driver, followed by operational convenience and peer effects. The change-shortage problem also appears here as a motivation for adoption, linking this chapter directly to the revenue-loss analysis in Chapter 2.

17.3. Business Duration and Adoption Motives

Business duration varies across motivation groups.

Table 17.2. Business duration by primary motivating factor

Motivating factor	<i>n</i>	Mean duration	Median duration
Customers kept asking for it	68	20.71	17.5
To avoid the problem of change	26	17.96	7.5
Other vendors were using it	21	16.81	7.0

Long-established vendors are especially likely to cite customer demand. This suggests a reactive form of adoption: as customer behaviour changes, even older businesses adapt. Newer vendors appear somewhat more likely to cite operational convenience or peer behaviour, but the differences are not statistically strong.

A Chi-square test was conducted after cleaning invalid entries and expanding multiple-response answers. The retained motivation categories were customer demand, other vendors using UPI, and avoiding the change problem. Shop age was grouped into six duration bands. The result was

$$\chi^2 = 9.688, \quad df = 10, \quad p = 0.468.$$

Since $p > 0.05$, there is no statistically significant association between shop age and the reason for adopting UPI. In practical terms, customer demand is widespread enough that it cuts across shop-age categories.

17.4. UPI Fee Arrangements

The fee-structure outcome is defined as

$$Y_i = \text{upi_rent}_i \in \{\text{No rent, Device rent, Per-sale rent}\}.$$

The cleaned distribution is shown below.

Table 17.3. Distribution of UPI fee arrangements

Fee arrangement	<i>n</i>	Percent	Valid percent
No rent	58	50.0%	53.7%
Device rent	37	31.9%	34.3%
Per-sale rent	13	11.2%	12.0%
Missing	8	6.9%	—

Most vendors either pay no rent or pay a device rent. Per-sale rent is less common.

17.5. Fee Arrangement and Business Trajectory

A Chi-square test was used to examine whether UPI fee arrangement is associated with reported business trajectory.

Table 17.4. UPI fee arrangement and business trajectory

Rent type	Grown	Same	Declined	Total
No rent	13	24	22	59
Device rent	7	24	13	44
Per-sale rent	3	6	4	13
Total	23	54	39	116

The test result was

$$\chi^2 = 2.081, \quad df = 4, \quad p = 0.7208.$$

There is no statistically significant evidence that fee arrangement is related to whether the business has grown, remained the same, or declined. Some expected cell counts in the per-sale rent category are below 5, so this result should be read cautiously. Nevertheless, the p-value is large enough that the table does not suggest a strong relationship.

17.6. Pre-model Screening for Fee Arrangement

Before estimating the multinomial model, several candidate predictors were screened against the fee-arrangement outcome.

Table 17.5. Association screening for UPI fee arrangement

Variable	Test	p-value	Cramer's V	Interpretation
Primary mode	Fisher's exact	0.66270	0.111	Weak
Goods type	Chi-square	0.07226	0.221	Weak
Age group	Fisher's exact	0.27630	0.178	Weak
Business duration group	Fisher's exact	0.03341	0.217	Weak but significant
UPI acceptance	Fisher's exact	0.51120	0.157	Weak
Ticket-size group	Fisher's exact	0.05305	0.199	Borderline

Business duration is significant at the 5% level, while ticket size is borderline. Goods type shows only a weak association and is not retained in the preferred model.

17.7. Age and UPI Adoption Status

Although UPI acceptance is highly imbalanced in the dataset, a descriptive comparison between UPI users and non-users is still informative. UPI users have median age category 2, corresponding to 30–44 years. Non-UPI users have median age category 3, corresponding to 45–59 years, and none of the non-UPI users are in the 18–29 age group.

A Mann–Whitney test comparing age distributions gives

$$U = 385.5, \quad p = 0.1310.$$

The difference is not statistically significant, mainly because the non-UPI group contains only 9 vendors. The descriptive pattern nevertheless points in the expected direction: non-users appear somewhat older, but the sample is too small to make a strong inferential claim.

17.8. UPI Adoption Utility

This section evaluates whether the net utility of UPI adoption is positive for street vendors. The main goal is to determine whether the operational and financial benefits of UPI adoption outweigh its associated costs, and whether this conclusion holds for both vendors who adopted UPI due to customer pressure and those who adopted it more proactively.

To assess overall utility, a composite *Net Benefit Score* (NB) was constructed from seven survey-based variables capturing business growth, transaction speed, operational ease, payment preference, fee burden, customer loss, and network reliability. Positive components were assigned to beneficial outcomes, while negative components were assigned to cost-related or disruptive outcomes. The theoretical range of NB is $[-3.0, 4.0]$, where $NB > 0$ indicates a net positive utility. The resulting score was then compared across two adoption-motivation groups:

- vendors who adopted UPI because customers kept asking for it;
- vendors who adopted UPI for other reasons, such as convenience or self-initiated adoption.

For each group, a one-sample t -test was used to test whether the mean net benefit was greater than zero:

$$H_0 : \mu_{NB} \leq 0 \quad \text{vs.} \quad H_1 : \mu_{NB} > 0.$$

Sample group	n	Mean NB	p -value
With customer pressure	58	0.9328	3.03×10^{-6}
Without customer pressure	31	1.1419	3.34×10^{-4}

Table 17.6. One-sample t -test results for the Net Benefit Score.

The null hypothesis is rejected for both groups, indicating that the net benefit of UPI adoption is significantly positive. In both cases, the observed benefits exceed the costs associated with rent, network issues, and lost sales. The mean net benefit is also higher for proactive adopters than for vendors who adopted under customer pressure, suggesting that voluntary adoption is associated with greater perceived operational efficiency.

The analysis shows that UPI adoption yields a positive net utility for street vendors. Although customer pressure is an important driver of adoption, the benefit remains statistically significant even among proactive adopters. Overall, the findings support the view that UPI provides a net operational advantage rather than merely adding a financial burden.

17.9. Overall Conclusion

The motivation analysis shows that UPI adoption is primarily demand-led. Customers asking for UPI is by far the most common reason for adoption, followed by avoiding the change problem and peer adoption by other vendors. The age of the shop does not significantly alter these motivations.

The fee-arrangement analysis shows that most vendors operate with no rent or device rent, while per-sale rent is comparatively uncommon. Fee arrangement is not significantly associated with business trajectory, but it is related to business duration and, to a lesser extent, ticket size. The strongest structural insight is that very mature businesses are more likely to operate under fee-based UPI arrangements, while larger-ticket vendors are less likely to pay device rent.

Taken together, the evidence suggests that UPI adoption among street vendors is less a purely voluntary technological upgrade and more a response to market pressure, customer expectations, and the practical need to avoid cash frictions.

Part V

Conclusions

Chapter 18

Conclusions

An extensive exploratory and descriptive analysis was carried out to understand the structural characteristics of the surveyed street-vendor ecosystem and to identify the major transactional and behavioural patterns present in the data. The EDA revealed a highly male-dominated vendor population (91.5%), with nearly three-fourths of vendors concentrated in the 30-59 age bracket. Operationally, the market exhibited strong geographic stability, with approximately 89.9% of vendors operating through permanent stalls rather than mobile setups. On the digital-payment side, UPI adoption itself was nearly universal, with 93% of vendors already accepting UPI, though the intensity of usage varied substantially across vendors. Exploratory plots further showed that most vendors operated in moderate ticket-size ranges (primarily Rs. 50–500), customer traffic distributions were strongly right-skewed, and both revenue and customer-volume variables exhibited substantial heterogeneity and outlier behaviour. Behavioural analysis additionally suggested that vendors overwhelmingly perceived UPI as improving transaction speed and financial tracking, while network failures and privacy concerns were relatively limited.

In order to carry out a meaningful inferential analysis, at first it is essential to come up with a statistically stable estimate of *UPI Intensity*. It turns out that the naive short-term adoption proportions are not ideal measures of UPI intensity as they don't account for the stochastic variability inherent in recall-based survey data. Therefore we opted for the Bayesian approach and constructed weighted Bayesian estimator for UPI intensity. This estimator is an adaptive weighted average of the recall-based estimate and the short-run empirical proportion, with weights determined jointly by the prior reliability (determined by internal consistency check of the vendor) and the volume of observed transactions. This prevented extreme and unstable estimates in sparse observational settings and reduced the impact of inconsistent recall responses.

We defined a Bayesian shrinkage estimator called Bayesian revenue instability index that regularised weekly revenue ranges using vendors' own long-run stability perceptions, producing a measure that captured experienced economic instability rather than raw mechanical dispersion alone. Comparing this instability index with the well known *Parkinson's Volatility Estimator*, we find that our estimator does have a positive but imperfect relationship with the Parkinson's measure where the variation is not due to noise but due to the structural contribution of the perceived-stability prior. The resulting instability index showed strong internal validity, with median instability increasing monotonically from "Very Steady" vendors to "Very Unsteady" vendors.

The inferential results reveal that most socio-demographic characteristics traditionally associated with digital adoption were not statistically significant once the stabilised Bayesian intensity estimator was used. Age group, mode of operation, business trajectory, type of goods sold, and business duration all failed to show significant association with UPI intensity at the 5% level. This suggests that many apparent adoption patterns observed in naive analyses are likely driven by short-term noise or measurement instability rather than genuine structural differences.

The analysis of cash change shortage established that this friction imposes a real and economically meaningful burden on vendors. The median estimated daily revenue loss due to inability to provide change was Rs. 240, corresponding to an annualised median loss of approximately Rs. 87,600. Although the distribution was highly skewed, the evidence clearly showed that intensified UPI usage substantially mitigates this problem. In the weighted logistic regression model, Bayesian UPI intensity had a strongly negative and statistically significant coefficient ($p = 0.0029$) with an odds ratio near 0.10.

Finally, the relationship between UPI intensity and revenue volatility turned out to be weaker and more heterogeneous than initially hypothesised. The exploratory analysis did not support a simple narrative that digital payments universally stabilise income. Instead, the effect appeared conditional on business structure and ticket size. Low-ticket vendors showed mild stabilising patterns, whereas medium- and high-ticket vendors exhibited volatility driven more by irregular demand than by payment frictions.

The analysis of vendor perception towards UPI revealed that digital payments are now viewed largely as a practical business tool rather than merely a technological alternative to cash. Most vendors reported that UPI improves transaction speed, simplifies financial tracking, and reduces operational friction in everyday transactions. At the same time, network-related issues and concerns regarding digital records were relatively limited, suggesting growing trust and behavioural integration of UPI in the street-vending economy.

Overall, the study demonstrates that digital-payment adoption among informal vendors should not be viewed as a binary phenomenon. The intensity, operational integration,

and transactional context of UPI usage matter far more than simple adoption itself. The findings also show the value of Bayesian regularisation techniques in informal-sector survey settings, where sparse observations, recall bias, and measurement instability can otherwise distort inference substantially.

Chapter 19

Acknowledgement

We would like to express our sincere gratitude to **Prof. Debasis Sengupta** and **Prof. Suman Majumdar** for their constant guidance and valuable suggestions throughout the course of this project. Their insightful discussions regarding various nuances of the methodology of data collection helped us a lot to carry out meaningful statistical analysis. We are truly thankful to all the vendors that kindly co-operated with us and provided us data to the best of their ability. Finally, we acknowledge the support of the **Indian Statistical Institute, Kolkata**, for providing the academic environment and resources necessary to carry out this work.

Part VI

Appendix

Appendix A

Appendix

Coding of Categorical variables

Table A.1. Coding of Categorical Variables

Variable	Categories	Code
Gender	Male	1
	Female	2
Age Group	18–29	1
	30–44	2
	45–59	3
	60+	4
Type of Goods Sold	Perishable	0
	Non-perishable	1
Primary Mode of Operation	Permanent Stall	0
	Mobile Cart	1
	Pavement	2
UPI Acceptance	No	0
	Yes	1
UPI Rent	No	0
	Yes (rent for device)	1

continued on next page

Variable	Categories	Code
	Yes (rent per sale)	2
Network Issue Frequency	Daily	0
	Weekly	1
	Monthly	2
	Rarely	3
Daily Amount Variation	Very Steady	0
	Somewhat Fluctuating	1
	Very Unsteady	2
UPI or Cash Preference in Transactions	UPI	0
	Cash	1
	No Difference	2
UPI Saves Time	Yes	0
	No	1
	No Difference	2
Business Trajectory	Grown	0
	Declined	1
	Same	2
Transaction Time using UPI	Faster	0
	No Change	1
	Slower	2
Digital Record Worry	No	0
	Yes	1
	Neutral	2
Customers Lost due to UPI Issues	Never	0
	Sometimes	1
	Often	2
Mode Preference	UPI	0
	Cash	1

continued on next page

Variable	Categories	Code
	Neutral	2
UPI and Cash Buckets Sum to Approximately 10?	Yes	1
	No	0
Very Steady: Busiest–Slowest Day Difference Less Than 30%?	Yes	1
	No (potential perception bias)	0
UPI Users in Last Hour Fall Within Daily UPI Bucket?	Major Contradiction	0
	Slightly Different	1
	Matches Exactly	2
QR Code Visibility	None Found	0
	Visible	1
	Hidden / Asked For	2
Atmosphere During Survey	Reluctant	0
	Suspicious	1
	Distracted	2
	Incentive Cooperation	4
	Cooperative	5